



**WL : ZIGBEE TRAINER
EXPERIMENTAL MANUAL**

.... A MESSAGE FROM

Today's system designers are faced with tomorrow's problems. BASIC ELECTRONICS is one of the important subject need to teach while learning electronics.

It is our vision to provide you with the product you need for training ensuring lasting reliability & quality.

OUR MOTTO;

- Light years ahead – refers to leadership.

As leaders in our industry in India, We are totally committed to servicing as the standard against which all are measured in the areas of

• Design • Quality • Value • Delivery • Support.

We are truly light years ahead of our competition in this area. That means that you our valued customers are guaranteed satisfaction.

As you will move through this manual you will quickly discover that We have a complete, truly innovative & superior training products we are so committed to quality that we back our products with a complete comprehensive warranty.

AKADEMIKA LAB SOLUTIONS

Sr. No. 15/8/1, Unit No.9, Kruti Industrial Estate, Opp.
Sangam Press, Kothrud, Pune, Maharashtra- 411038

INDIA

Contact No.:+91-9004904462

+91-7447438443

Email: lab@akademika.in

www.akademika.in

SAFETY RULES

- Read carefully and follow the instructions mentioned in this manual. This user manual includes all the important points about the installation, use and the maintenance of the product. Keep this manual always with you, for quick reference.
- After unpacking the product, arrange all the accessories in proper order, so that their integrity is checked with the packing list. Also, ensure that the accessories have no visible damage.
- Before connecting the power supply to the kit, be sure that the jumpers and the connecting chords are connected correctly, as per the experiment.
- This kit must be employed only for the use for which it has been conceived, i.e. as educational kit and must be used under the direct survey of expert personnel. Any other use is inadvisable and dangerous too. The manufacturer cannot be considered responsible for eventual damages due to improper, wrong or unreasonable uses.
- In case of any fault or malfunctioning in the trainer kit, turn off the power supply. Please do not tamper the kit. If you require our service, kindly contact the service centre for technical assistance.
- The kits are liable to malfunction/under-perform if they are not operated under standard environmental conditions of temperature and humidity.

WARRANTY

This kit is warranted against defects in workmanship and materials. Any failure due to defect in either workmanship or material should occur under normal use within a year from the original date of purchase, such failure will be corrected free of charge to the purchaser by repair or replacement of defective part or parts. When the failure is result of user's neglect, natural disaster or accident, we would charge for repairs, regardless of the warranty period. The warranty does not cover include perishable items like connecting chords, crystals, etc. and other imported items.

Conditions and Limitations

The warranty is void and inapplicable if the defective product is not brought or sent to our authorized service center or sales outlet within the warranty period. Defective product will be Akademika Lab Solutions sole judgment. The defective product will be replaced with a new one or repaired, without charge or with charge.

In the warranty period if the service is needed, the purchaser should get in touch with the service center or the sales outlet. The purchaser should return the product to the service center or the sales outlet at his or her sole expense. The loss and damage in transit will be outside the preview of this warranty. A returned product must be accompanied by a written description of the defects. Type and Model No. of the kit has to be mentioned specifically. We return the product to the purchaser at our expense. In case the warranty does not cover the product on Akademika Lab Solutions judgment, we would repair the product after obtaining prior permission from purchaser who would receive an estimate statement about the repairing charges. In such cases, Akademika Lab Solutions bares the transporting expenses required to send back all the repaired products for the moment, and then repairs and transporting expenses will be charged against the purchaser by the statement of accounts.

When the authorized sales agents sell our products, they must notify the purchaser of the warranty contents, but they have no rights to stretch the meaning of original warranty contents or to offer an additional warranty. Akademika Lab Solutions does not provide any other promise or suggestive warranty and hold no liability for the damage caused by negligence, abnormal use or natural disaster. Akademika Lab Solutions is not responsible for the damages even if it is notified of above dangers in advance as well.

For more special service or overall repairs, maintenance and up gradation of products, be sure to contact our service center or the sales outlet.

INDEX

SR. NO	CHAPTER	PAGE NO
1.	Introduction	06
2.	Technical Specifications	07
3.	Functional Block Diagram	08
4.	Circuit Diagram	09
5.	Study of WIRELESS COMMUNICATION SYSTEM	11
6.	Study of ZigBee TECHNOLOGY	18
7.	Study of ZigBee TOPOLOGIES & PROTOCOL	23
8.	Study of ZigBee MODULE & AT COMMANDS	27
9.	ZigBEE Trainer Hardware and Software Details	49

List of Experiments

10.	Experiment No. 1	
	CONFIGURATION OF THE ZIGBEE MODULE	
	A. Configuring ZigBee as Coordinator	60
	B. Configuring ZigBee as Router	64
	C. Configuring ZigBee as END DEVICE	67
11.	Experiment No. 2	71
	CONFIGURATION WITH TWO ZIGBEE MODULE	
	A. Data Comm: Coordinator Module to Router Module	80
	B. Data Comm: Coordinator Module to End device	82

12. Experiment No. 3**INTERFACING THE PHERIPHERALS**

A. Interfacing on board Peripherals	87
• Interfacing with LEDs and SWITCHES	90
• Interfacing with RELAY & BUZZER	91
• Interfacing with ADC using Variable Resistor	92
• Interfacing with LDR	93
• Interfacing with TEMPERATURE SENSOR	94
B. Interfacing remote board Peripherals	95
• Interfacing with LEDs and SWITCHES	95
• Interfacing with RELAY & BUZZER	98
• Interfacing with ADC using Variable Resistor	100
• Interfacing with LDR	102
• Interfacing with TEMPERATURE SENSOR	104

INTRODUCTION

ZigBee is a specification for a suite of high level communication protocols used to create personal area networks built from small, low-power digital radios. ZigBee is based on an IEEE 802.15 standard. Though low-powered, ZigBee devices can transmit data over long distances by passing data through intermediate devices to reach more distant ones, creating a mesh network.

ZigBee is used in applications that require only a low data rate, long battery life, and secure networking. ZigBee has a defined rate of 250 Kbit/s, best suited for periodic or intermittent data or a single signal transmission from a sensor or input device. ZigBee was conceived in 1998, standardized in 2003, and revised in 2006. The name refers to the waggle dance of honey bees after their return to the beehive.

The ZigBee Trainer is designed to study the Wireless Technology and IEEE 802.15 Standard. This trainer covers all ZigBEE Topologies & Protocol. With the help of this trainer user can simulate / observe various input and output devices. Also user can select various I/O standard interfaces to the ZigBee Module.

The ZigBee Trainer include ProBee-ZE10 ZigBee module from SENA Company. It is an all-in-one ZigBee OEM module with integrated ZigBee core and radio/antenna circuits. The ZE10 Module is designed to meet specific requirements of low cost and low power wireless applications such as Home Automation, Smart Energy, Health care, Building Automation and Telecom Services by adopting the ZigBee technology. The ZE10 can transfer data at up to 250 kbps and reach up to 1.6 km (1 mile) outdoor line-of-sight.

Trainer incorporate on board peripherals such as LEDs, Switches, Relay, Buzzer and Sensors (LCR, TEMP: Sensor). This help to the user to understand how to communicate different peripherals to ZigBee Module.

TECHNICAL SPECIFICATIONS

FEATURES

- ZigBee Transceiver (Pro Bee-ZE10)
- USB Interface
- ZigBee Status Indicator
- User Interface Switches & LEDs
- On Board Peripheral (Relay & Buzzer)
- Sensors to measure Temperature & Light

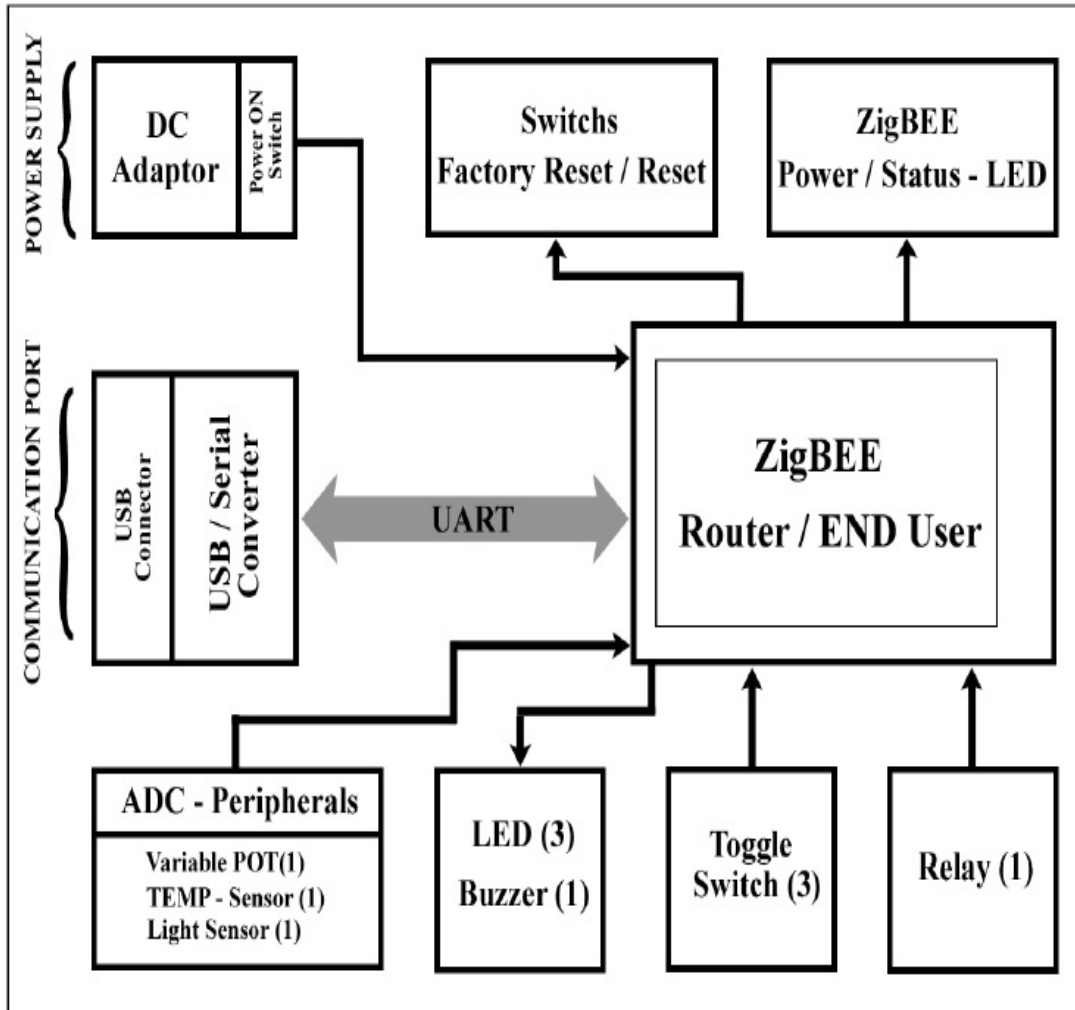
SPECIFICATION

- Integrated 2.4GHz, IEEE 802.15.4-Compliant ZigBee Transceiver
- Power Supply (+5V & +3V)
- On board USB to Serial Interface
- On board ZigBee Status Indicator.
- On Board Input Toggle Switches (3 NO) & Output LEDs (3)
- On Board 5V SPDT Relay and Buzzer
- On board Light & Temperature Sensor (LDR & LM 35)

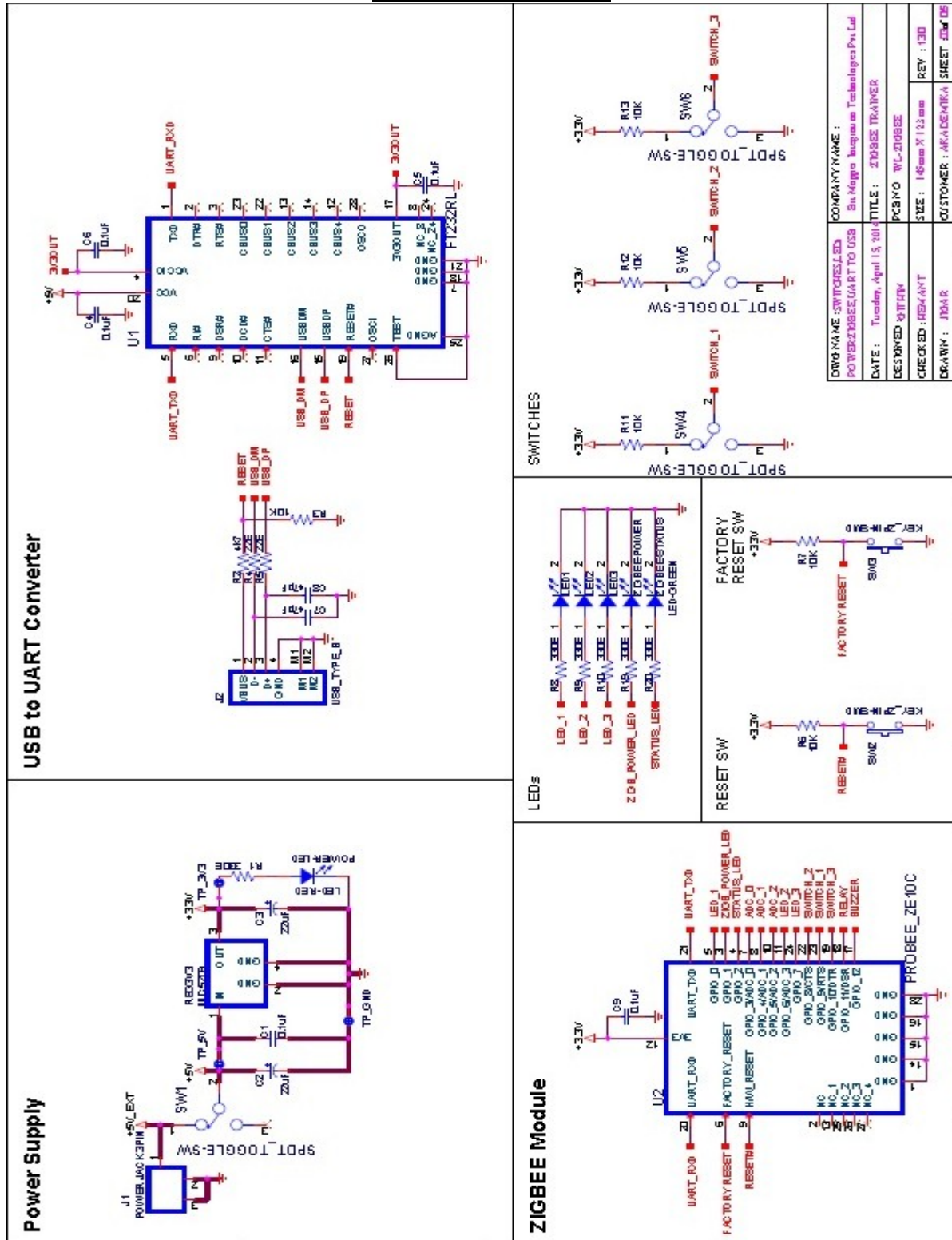
List of Deliverables

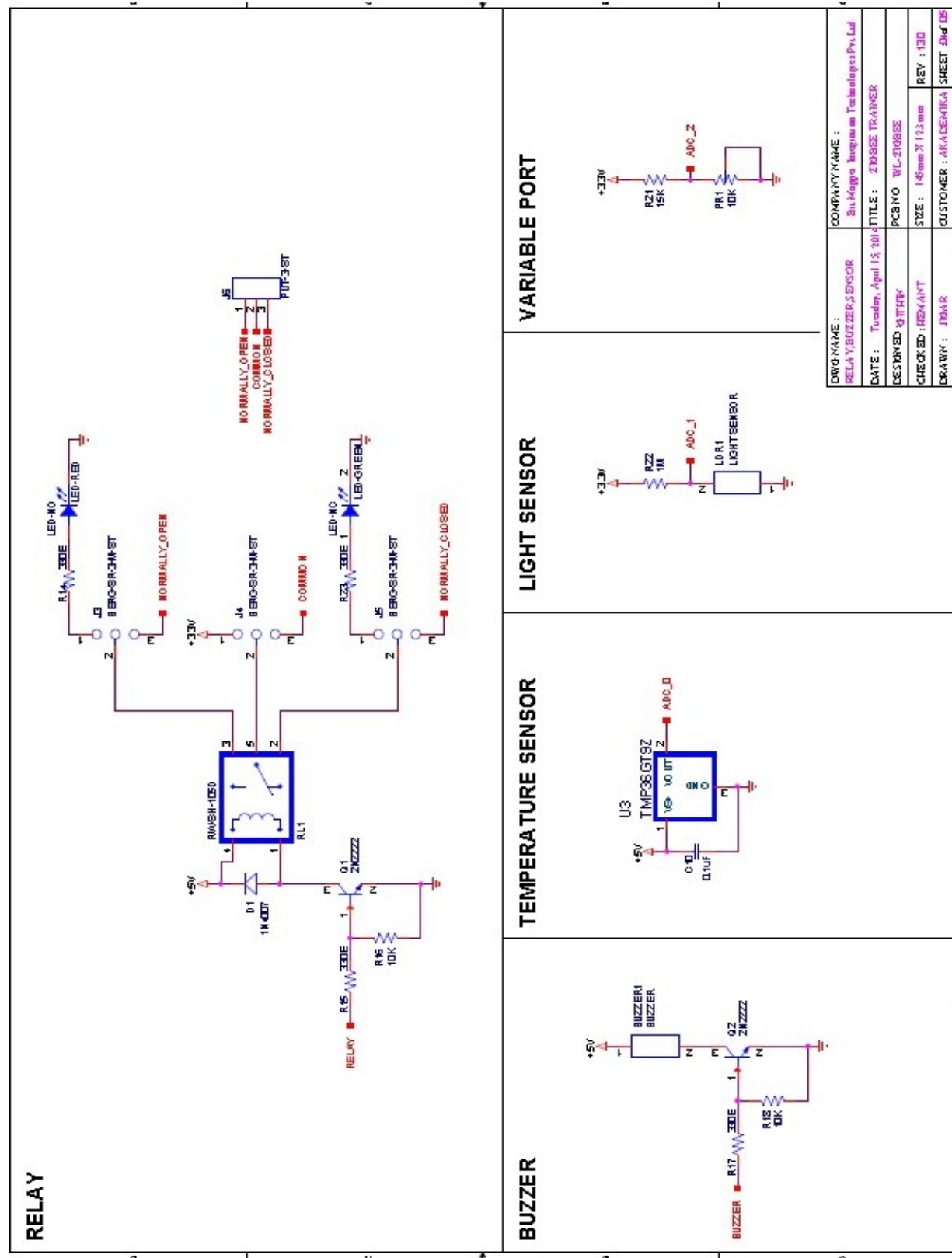
- ZigBee Trainer Kit : 02
- Power Supply : 02
- USB Cable : 02
- Software CD : 01
- Experimental Manual : 01

Functional Block Diagram



Circuit Diagram

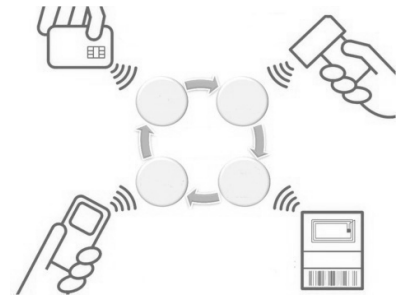




WIRELESS COMMUNICATION SYSTEM

What is a Wireless System?

A wireless system is any collection of elements that operate interdependently and use unguided electromagnetic-wave propagation to perform some specified function.



Some examples of systems that fit this definition are

- Systems that convey information between two or more locations, such as personal communication systems, police and fire department radio systems, commercial broadcast systems, satellite broadcast systems, telemetry and remote monitoring systems.
- Systems that sense the environment or objects in the environment, including radar systems that may be used for detecting the presence of objects in some region or volume of the environment and measuring their relative motion and position, systems for sensing or measuring atmospheric conditions and systems for mapping the surface of the Earth or planets
- Systems that aid in navigation or determine the location of an object on the Earth or in space

Each of these systems contains at least one transmitting antenna and at least one receiving antenna. In the abstract, an antenna may be thought of as any device that converts a guided signal, such as a signal in an electrical circuit or transmission line, into an unguided signal propagating in space, or vice versa. For example, the WiFi local area network computer interface uses a single antenna that is switched between transmitter and receiver.

As the examples show, some systems may be used to convey information, whereas others may be used to extract information about the environment based on how the transmitted signal travels the path between transmitting and receiving antennas. In either case, the physical and electromagnetic environment in the neighborhood of the path may significantly modify the signal. We define a channel as the physical and electromagnetic environment surrounding and connecting the endpoints of the transmission path that is, surrounding and connecting the system's transmitter and receiver. *(A channel may consist of wires, waveguide and coaxial cable, fiber, the Earth's atmosphere and surface, free space, and so on.)* When a wireless system is used to convey information between endpoints, the environment often corrupts the signal in an unpredictable way and impairs the system's ability to extract the transmitted information accurately at a receiving end.

WIRELESS COMMUNICATIONS

It is a type of data communication that is performed and delivered wirelessly. This is a broad term that incorporates all procedures and forms of connecting and communicating between two or more devices using a wireless signal through wireless communication technologies and devices. Wireless communication generally works through electromagnetic signals that are broadcast by an enabled device within the air, physical environment or atmosphere. The sending device can be a sender or an intermediate device with the ability to propagate wireless signals. The communication between two devices occurs when the destination or receiving intermediate device captures these signals, creating a wireless communication bridge between the sender and receiver device. Wireless communication has various forms, technology and delivery methods.

The first wireless transmitters went on the air in the early 20th century using radiotelegraphy. Later, as modulation made it possible to transmit voices and music via wireless, the medium came to be called "RADIO". With the advent of television, fax, data communication, and the effective use of a larger portion of the spectrum, the term "wireless" has been resurrected.



Common examples of wireless equipment in use today include:

Cellular phones and pagers -- provide connectivity for portable and mobile applications, both personal and business.

Global Positioning System (GPS) -- allows drivers of cars and trucks, captains of boats and ships, and pilots of aircraft to ascertain their location anywhere on earth.

Cordless computer peripherals -- the cordless mouse is a common example; keyboards and printers can also be linked to a computer via wireless.

Cordless telephone sets -- these are limited-range devices, not to be confused with cell phones.

Home-entertainment-system control boxes -- the VCR control and the TV channel control are the most common examples; some hi-fi sound systems and FM broadcast receivers also use this technology.

Two-way radios -- this includes Amateur and Citizens Radio Service, as well as business, marine, and military communications.

Baby monitors -- these devices are simplified radio transmitter/receiver units with limited range.

Satellite television -- allows viewers in almost any location to select from hundreds of channels.

Wireless LAN Or Local Area Networks -- provide flexibility and reliability for business computer users.

Wireless technology is rapidly evolving, and is playing an increasing role in the lives of people throughout the world. In addition, ever-larger numbers of people are relying on the technology directly or indirectly. More specialized and exotic examples of wireless communications and control include:

Global System for Mobile Communication (GSM) -- a digital mobile telephone system used in Europe and other parts of the world

General Packet Radio Service (GPRS) -- a packet-based wireless communication service that provides continuous connection to the Internet for mobile phone and computer users.

Enhanced Data GSM Environment (EDGE) -- a faster version of the Global System for Mobile (GSM) wireless service

Universal Mobile Telecommunications System (UMTS) -- a broadband, packet-based system offering a consistent set of services to mobile computer and phone users no matter where they are located in the world

Wireless Application Protocol (WAP) -- a set of communication protocols to standardize the way that wireless devices, such as cellular telephones and radio transceivers, can be used for Internet access.

i-Mode -- the world's first "smart phone" for Web browsing, first introduced in Japan; provides color and video over telephone sets.

Wireless can be divided into:

FIXED WIRELESS -- the operation of wireless devices or systems in homes and offices, and in particular, equipment connected to the Internet via specialized modems.

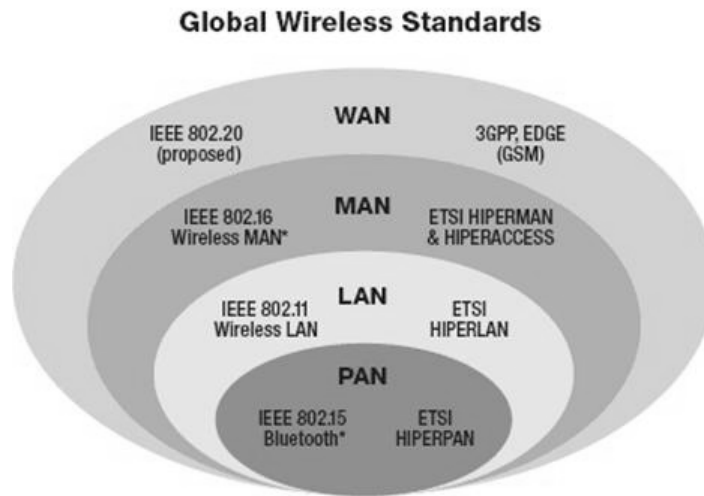
MOBILE WIRELESS -- the use of wireless devices or systems aboard motorized, moving vehicles; examples include the automotive cell phone and PCS (personal communications services).

PORTABLE WIRELESS -- the operation of autonomous, battery-powered wireless devices or systems outside the office, home, or vehicle; examples include handheld cell phones and PCS units.

IR WIRELESS -- the use of devices that convey data via IR (infrared) radiation; employed in certain limited-range communications and control systems

WIRELESS STANDARDS

Different methods and standards of wireless communication have developed across the world, based on various commercially driven requirements. These technologies can roughly be classified into four individual categories, based on their specific application and transmission range. These categories are summarized in the figure below



PERSONAL AREA NETWORK (PAN)

A Personal Area Network (PAN) is a computer network used for communication among computer devices (including telephones and personal digital assistants) close to one person. The reach of a PAN is typically a few meters. PAN's can be used for communication among the personal devices themselves (intrapersonal communication), or for connecting to a higher level network and the Internet.

Personal area networks may be wired with computer buses such as USB and FireWire. However, a Wireless Personal Area Network (WPAN) is made possible with network technologies such as Infrared (IrDA) and Bluetooth.



Bluetooth: Bluetooth is an industrial specification for wireless personal area networks (PANs), also known as IEEE 802.15.1. Bluetooth provides a way to connect and exchange information between devices such as personal digital assistants (PDAs), mobile phones, laptops, PCs, printers, digital cameras and video game consoles via a secure, globally unlicensed short-range radio frequency. Bluetooth is a radio standard and communications protocol primarily designed for low power consumption, with a short range (power class dependent: 1 meter, 10 meters, 100 meters) based around low-cost transceiver microchips in each device.

Infrared (IrDA): The Infrared Data Association (IrDA) defines physical specifications communications protocol standards for the short range exchange of data over infrared light, for typical use in Personal Area Networks.

LOCAL AREA NETWORK (LAN)

A wireless LAN or WLAN is a wireless Local Area Network, which is the linking of two or more computers without using wires. It uses radio communication to accomplish the same functionality that a wired LAN has.

WLAN utilizes spread-spectrum technology based on radio waves to enable communication between devices in a limited area, also known as the basic service set. This gives users the mobility to move around within a broad coverage area and still be connected to the network.



IEEE 802.11: IEEE 802.11, the Wi-Fi standard, denotes a set of Wireless LAN/WLAN standards developed by working group 11 of the IEEE LAN/MAN Standards Committee (IEEE 802). The 802.11 family currently includes six over-the-air modulation techniques that all use the same protocol. The most popular (and prolific) techniques are those defined by the b, a, and g amendments to the original standard.

The table below summarizes the different 802.11 standards:

Protocol	Release Date	Op. Frequency	Data Rate (Typical)	Data Rate (Max)	Range (Indoor)
Legacy	1997	2.4 -2.5 GHz	1 Mbit/s	2 Mbit/s	?
802.11a	1999	5.15-5.35/5.47-5.725/5.725-5.875 GHz	25 Mbit/s	54 Mbit/s	~30 meters (~100 feet)
802.11b	1999	2.4-2.5 GHz	6.5 Mbit/s	11 Mbit/s	~50 meters (~150 feet)
802.11g	2003	2.4-2.5 GHz	11 Mbit/s	54 Mbit/s	~30 meters (~100 feet)
802.11n	2006 (draft)	2.4 GHz or 5 GHz bands	200 Mbit/s	540 Mbit/s	~50 meters (~160 feet)

METROPOLITAN AREA NETWORK (MAN)

Wireless Metropolitan Area Network (MAN) is the name trademarked by the IEEE 802.16 Working Group on Broadband Wireless Access Standards for its wireless metropolitan area network standard (commercially known as WiMAX), which defines broadband Internet access from fixed or mobile devices via antennas. Subscriber stations communicate with base-stations that are connected to a core network. This is a good alternative to fixed line networks and it is simple to build and relatively inexpensive



WiMAX: it is defined as Worldwide Interoperability for Microwave Access by the WiMAX Forum, formed in June 2001 to promote conformance and interoperability of the IEEE 802.16 standard, officially known as Wireless MAN. The Forum describes WiMAX as "a standards-based technology enabling the delivery of last mile wireless broadband access as an alternative to cable and DSL".

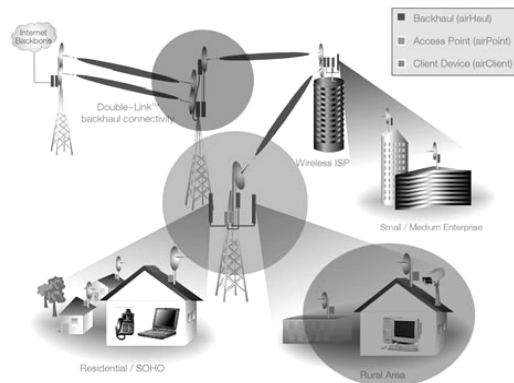
The range of WiMAX probably generates more confusion than any other single aspect of WiMAX. It is common to see statements in the media describing WiMAX multipoint coverage extending 30 miles. In a strict technical sense (in some spectrum ranges) this is correct, with even greater ranges being possible in point to point links. In practice in the real world (and especially in the license-free bands) this is wildly overstated especially where non line of sight (NLOS) reception is concerned.

WIDE AREA NETWORK (WAN)

A Wide Area Network or WAN is a computer network covering a broad geographical area. Contrast with personal area networks (PAN's), local area networks (LAN's) or metropolitan area networks (MAN's) that are usually limited to a room, building or campus. The largest and most well-known example of a WAN is the Internet.

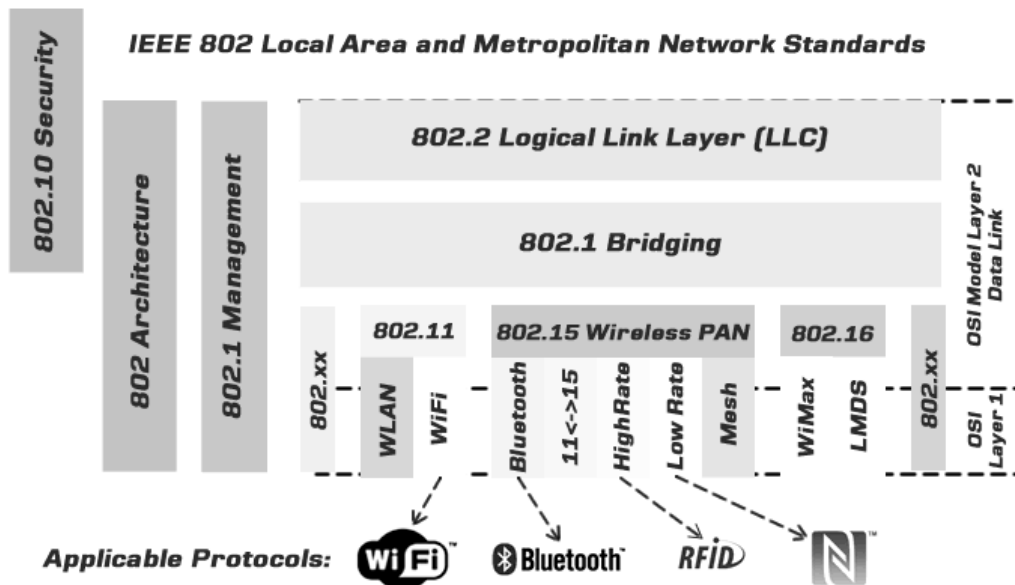
WAN's are used to connect local area networks (LAN's) together, so that users and computers in one location can communicate with users and computers in other locations. Many WAN's are built for one particular organization and are private. Others, built by Internet service providers, provide connections from an organization's LAN to the Internet.

In addition, WAN's also referred to Mobile Data Communications, such as GSM, GPRS and 3G. Please refer to our Mobility section for further details.



TECHNOLOGY SUMMARY

These wireless communication technologies evolved over time to enable the transmission of larger amounts of data at greater speeds across a global network. The following figure summarizes the technical details of each cluster of technologies.



ZigBee TECHNOLOGY

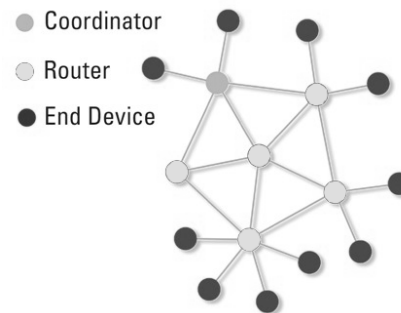
The explosion of wireless technologies in recent years has allowed the emergence of several standards, especially in the Industrial Scientific & Medical (ISM) band. Among these emerging standards ZigBee is considered one of the most. ZigBee is a low power spin off of Wi-Fi. It is a specification for small, low power radios based on IEEE 802.15.4 – 2003 Wireless Personal Area Networks standard. The specification was accepted and ratified by the ZigBee alliance in December 2004.

ZigBee Alliance is a group of more than 300 companies including industry majors like Philips, Mitsubishi Electric, Epson, Atmel, Texas Instruments etc. which are committed towards developing and promoting this standard. The alliance is responsible for publishing and maintaining the ZigBee specification and has updated it time and again after making it public for the first time in 2005. Most of the recent devices conform to the ZigBee 2007 specifications has two feature sets– ZigBee and ZigBee Pro. The manufacturers which are members of the Alliance provide software, hardware and reference designs to anyone who wants to build applications using ZigBee.

ZIGBEE PLATFORM OVERVIEW

ZigBee Alliance is the creation of a reliable, low-cost, low-power, open global standard for low data rate wireless solutions, while allowing multi-hop routing of data. The ZigBee standard through mesh network capability and AES 128-bit encryption provides support for self-healing and high security.

Figure describes a ZigBee network topology which typically includes three types of devices or nodes:



COORDINATOR: One coordinator exists in each network. It starts the network and handles management functions as well as data routing functions. These functions require that the coordinator always be powered. Therefore, this type of node is recommended to be main-powered.

ROUTERS: In most cases, routers are also main-powered. They help carry data across multi-hop ZigBee networks including a variable number of routers and, in some cases, are without routers, thus, transforming the network into a point-to-multipoint.

END DEVICES: These are devices that are battery-powered due to their low-power consumption. They sleep most of the time and wake up regularly to collect and transmit data. Devices such as sensors are configured as end devices. They are connected to the network through the routers.

The type of node is assigned during the commissioning process. The main-powered requirement for coordinators can be a limiting factor for ZigBee, especially if minimizing power consumption is actively targeted for each and every device. As the number of nodes in a ZigBee network increases, potential communication bottlenecks can occur in some

parts of the network. Two main techniques can be used to limit the congestion issue within ZigBee networks:

- Through node placement and adequate ZigBee router coverage within an installation area. This provides multiple paths for messages to reach a concentrator and alleviates potential bottleneck points in the network.
- Use several data concentrators instead of a single concentrator. This can reduce the number of hops required for nodes for their messages to reach a concentrator, and also reduces the chance of having a single point of failure.

In addition to the three methods mentioned earlier, techniques such as network partitioning through channel or PANID (or network ID) are available to deal with the ZigBee network congestion.

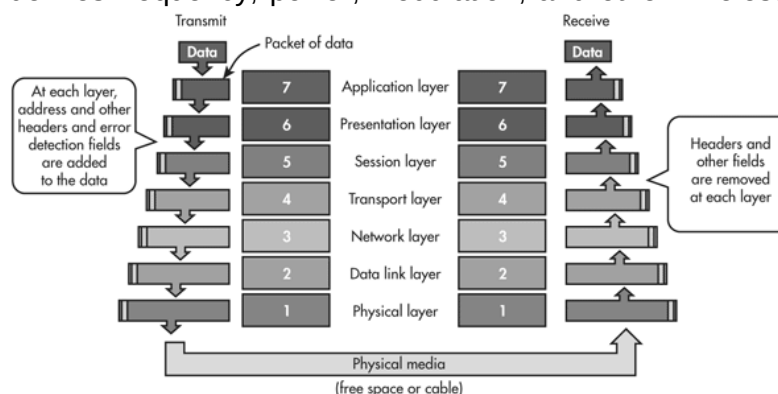
IEEE 802.15.4 and ZigBee BASICS

The Institute of Electrical and Electronics Engineers (IEEE) supports many working groups to develop and maintain wireless and wired communications standards. For example, 802.3 is wired Ethernet and 802.11 is for wireless LANs (WLANs), also known as Wi-Fi. The 802.15 group of standards specifies a variety of wireless personal area networks (WPANs) for different applications. For instance, 802.15.1 is Bluetooth, 802.15.3 is a high-data-rate category for ultra-wideband (UWB) technologies, and 802.15.6 is for body area networks (BAN). There are several others.

The 802.15.4 category is probably the largest standard for low-data-rate WPANs. It has many subcategories. The 802.15.4 category was developed for low-data-rate monitor and control applications and extended-life low-power-consumption uses. The basic standard with the most recent updates and enhancements is 802.15.4a/b, with 802.15.4c for China, 802.15.4d for Japan, 802.15.4e for industrial applications, 802.15.4f for active (battery powered) radio-frequency identification (RFID) uses, and 802.15.4g for smart utility networks (SUNs) for monitoring the Smart Grid. All of these special versions use the same base radio technology and protocol as defined in 802.15.4a/b.

The 802.15.4 standard defines the physical layer (PHY) and media access control (MAC) layer of the Open Systems Interconnection (OSI) model of network operation. The PHY defines frequency, power, modulation, and other wireless conditions of the link. The MAC

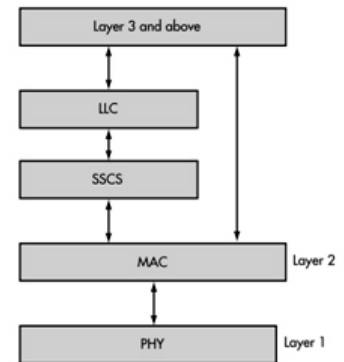
defines the format of the data handling. The remaining layers define other measures for handling the data and related protocol enhancements including the final application.



Most networking systems, both wired and wireless use the OSI communications model. Most systems also use at least the first four layers, but many do not use all seven layers. More specifically, Figure 2 shows the layer 1 and layer 2 details of 802.15.4.

The 802.15.4 standard uses only the first two layers plus the logical link control (LLC) and service specific convergence sub-layer (SSCS) additions to communicate with all upper layers as defined by additional standards.

The goal of the standard is to provide a base format to which other protocols and features could be added by way of the upper layers (layers 3 through 7). While three frequency assignments are available, the 2.4-GHz band is by far the most widely used. Most available chips and modules use this popular ISM band.

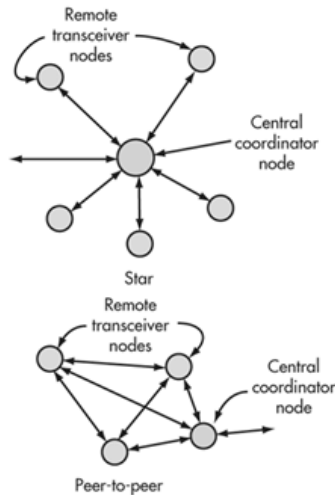


OPTIONS FOR FREQUENCY ASSIGNMENTS			
Geographical regions	Europe	Americas	Worldwide
Frequency assignment	868 to 868.6 MHz	902 to 928 MHz	2.4 to 2.4835 GHz
Number of channels	1	10	16
Channel bandwidth	600 kHz	2 MHz	5 MHz
Symbol rate	20 ksymbols/s	40 ksymbols/s	62.5 ksymbols/s
Data rate	20 kbits/s	40 kbits/s	250 kbits/s
Modulation	BPSK	BPSK	Q-QPSK

The standard uses direct sequence spread spectrum (DSSS) modulation. It is highly tolerant of noise and interference and offers coding gain to improve link reliability. Standard binary phase-shift keying (BPSK) is used in the two low-speed versions, while offset-quadrature phase-shift keying (O-QPSK) is used for the higher-data-rate version. O-QPSK has a constant wave envelope meaning that more efficient non-linear power amplification techniques can be used to minimize power consumption.

With regard to channel access, 802.15.4 uses carrier sense multiple access with collision avoidance (CSMA-CA). This multiplexing approach lets multiple users or nodes access the same channel at different times without interference. Most transmissions are short packets that occur infrequently for a very low duty cycle (<1 %), minimizing power consumption. The minimum power level defined is -3 dBm or 0.5 mW. Most modules use 0 dBm or 1 mW. However, some 20-dBm or 100-mW modules are available.

Transmission range varies considerably depending on the nature of the path that must for the most part be line of sight (LOS). Transmit power level and receiver sensitivity is also factors. Under the best conditions the range can be as great as 1000 meters with a clear outdoor path. Most applications cover a shorter range of 10 to 75 meters

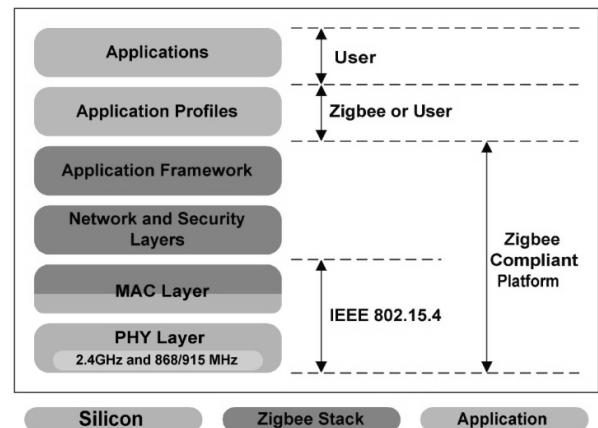


With regard to networking capability, 802.15.4 defines two topologies. One of them is a basic star. All communications between nodes must pass through the central coordinator node. A basic peer-to-peer (P2P) topology is also defined. Any device may then talk to any other device. This basic topology may be expanded into other topologies in the upper network layers, such as the popular mesh topology.

The 802.15.4 standard defines the star (a) and peer-to-peer (b) common network topologies.

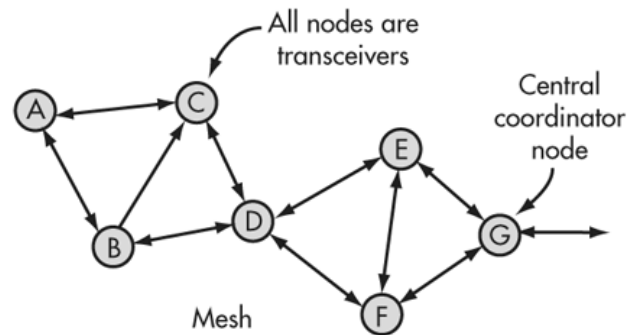
ZigBee

The most widely deployed enhancement to the 802.15.4 standard is ZigBee, which is a standard of the ZigBee Alliance. The organization maintains, supports, and develops more sophisticated protocols for advanced applications. It uses layers 3 and 4 to define additional communications features (Fig. 4). These enhancements include authentication with valid nodes, encryption for security, and a data routing and forwarding capability that enables mesh networking. The most popular use of ZigBee is wireless sensor networks using the mesh topology. The ZigBee protocol is defined by layer 3 and above. It works with the 802.15.4 layers 1 and 2.



The main benefit of the mesh topology is that any node can communicate with any other node, if not directly if within range, but indirectly by relaying the transmission through multiple additional nodes. The network then can spread out over a larger area. Furthermore, it increases network reliability as it still functions even if one node is disabled. There are usually alternate paths through the network to sustain a connection.

For example, if node A wishes to communicate with node G, it can relay data through nodes C and E. If node C fails, another path is via nodes B, D, and F. ZigBee mesh networks are self-configuring and self-healing.



In a mesh network, each node communicates with its closest neighbor as conditions permit. Note that there are alternate paths between any two nodes. ZigBee is also available in a version that supports energy harvesting where no battery or ac mains power is available. And, one of the key benefits of ZigBee is the availability of pre-developed applications. These upper-layer software additions implement specialized uses for ZigBee.

Some of these applications include:

- Building automation for commercial monitoring and control of facilities
- Remote control (RF4CE or RF for consumer electronics)
- Smart energy for home energy monitoring
- Health care for medical and fitness monitoring
- Home automation for control of smart homes
- Input devices for keyboards, mice, touch pads, wands, etc.
- Light Link for control of LED lighting
- Retail services for shopping related uses
- Telecom services
- Network services related to large mesh networks

The ZigBee Alliance also offers full testing and certification of ZigBee-enabled products to ensure interoperability. ZigBee has been around for more than 10 years now and is widely used. It is a great option for many applications.

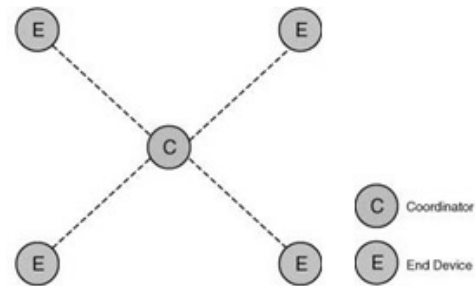
ZIGBEE TOPOLOGIES & PROTOCOL ARCHITECTURE

TOPOLOGIES

ZigBee uses the IEEE 802.15.4 2003 specification for its physical layer and MAC layer. IEEE 802.15.4 offers STAR, TREE, CLUSTER TREE, and MESH Topologies; however, ZigBee supports only STAR, TREE, and MESH TOPOLOGIES. It uses an association hierarchy; a Coordinator is must to make a ZigBee network. The device joining to the network can either be a Router or an End Device, and Routers can accept more devices.

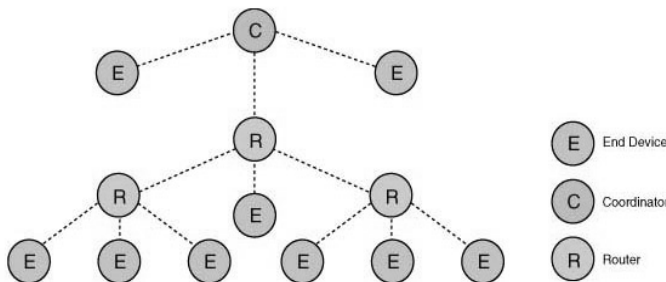
STAR TOPOLOGY:

The star topology consists of a coordinator and several end devices (nodes), as shown in Figure. In this topology, the end device communicates only with the coordinator. Any packet exchange between end devices must go through the coordinator. The disadvantage of this topology is the operation of the network depends on the coordinator of the network, and because all packets between devices must go through coordinator, the coordinator may become bottlenecked. Also, there is no alternative path from the source to the destination. The advantage of star topology is that it is simple and packets go through at most two hops to reach their destination.



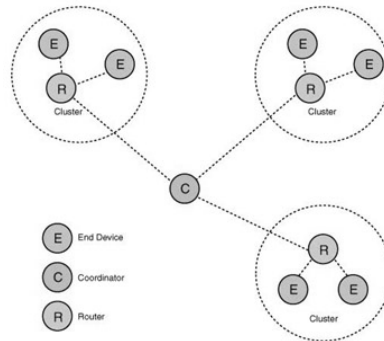
TREE TOPOLOGY:

In this topology, the network consists of a central node (root tree), which is a coordinator, several routers, and end devices, as shown in Figure. The function of the router is to extend the network coverage. The end nodes that are connected to the coordinator or the routers are called children. Only routers and the coordinator can have children. Each end device is only able to communicate with its parent (router or coordinator). The coordinator and routers can have children and, therefore, are the only devices that can be parents. An end device cannot have children and, therefore, may not be a parent. A special case of tree topology is called a *cluster tree topology*.



The disadvantages of tree topology are

- If one of the parents becomes disabled, the children of the disabled parent cannot communicate with other devices in the network.
- Even if two nodes are geographically close to each other, they cannot communicate directly.

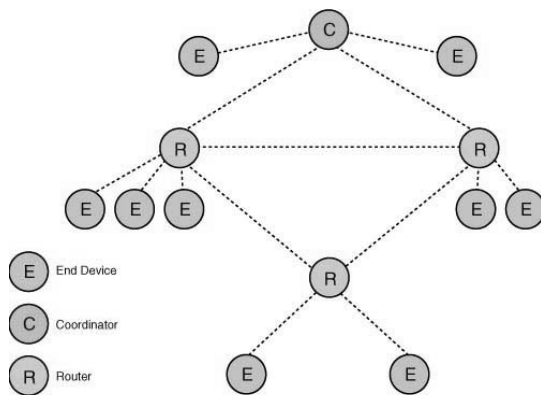
CLUSTER TREE TOPOLOGY:

A cluster tree topology is a special case of tree topology in which a parent with its children is called a cluster, as shown in Figure 2.4. Each cluster is identified by a cluster ID. ZigBee does not support cluster tree topology, but IEEE 802.15.4 does support it.

MESH TOPOLOGY:

Mesh topology, also referred to as a peer-to-peer network and consists of one coordinator, several routers, and end devices, as shown in Figure.

The following are the characteristics of a mesh topology:



A mesh topology is a multihop network; packets pass through multiple hops to reach their destination.

The range of a network can be increased by adding more devices to the network.

It can eliminate dead zones.

A mesh topology is self-healing, meaning during transmission, if a path fails, the node will find an alternate path to the destination.

Devices can be close to each other so that they use less power.

Adding or removing a device is easy.

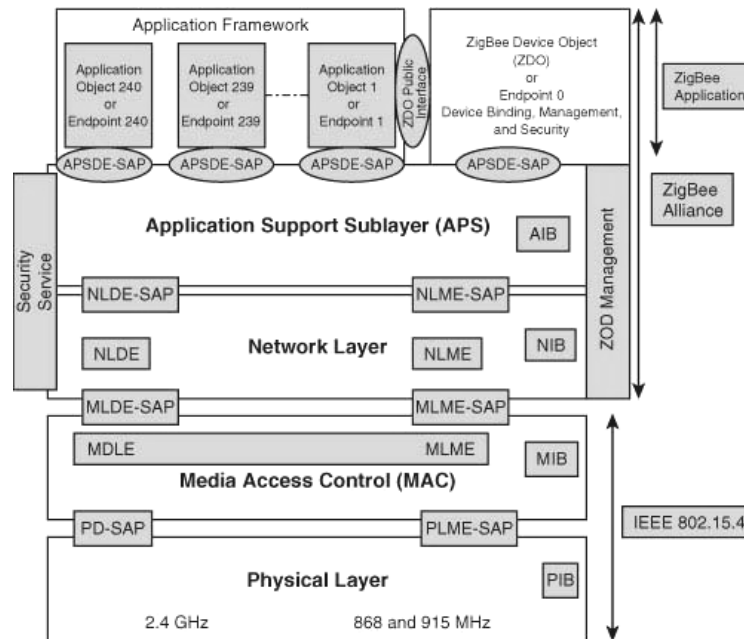
Any source device can communicate with any destination device in the network.

Compared with star topology, mesh topology requires greater overhead.

Mesh routing uses a more complex routing protocol than a star topology.

PROTOCOL ARCHITECTURE

The figure shows ZigBee protocol architecture. The ZigBee Alliance developed the ZigBee device object (ZDO), the application support sublayer (APS), the network layer, and security management. IEEE 802.15.4 is used for the MAC layer and physical layer.



The ZigBee protocol architecture is divided into three sections, as follows:

- IEEE 802.15.4, which consists of the MAC and physical layers.
- ZigBee layers, which consist of the network layer, the ZigBee device object (ZDO), the application sublayer, and security management.
- Manufacturer application: Manufacturers of ZigBee devices can use the ZigBee application profile or develop their own application profile.

PHYSICAL LAYER

The physical layer performs modulation on outgoing signals and demodulation on incoming signals. It transmits information and receives information from a source. Table 2.2 shows the physical layer frequency band, data rate, and channel numbers.

Physical Layer Frequency Band

Frequency Band	Country	Data Rate	Channel Numbers
868.3MHz	European countries	20Kbps	0
902–928 MHz	United States	40Kbps	1–10
2.405GHz	Worldwide	250Kbps	11–26

MEDIA ACCESS CONTROL (MAC) LAYER

The functions of the MAC layer are to access the network by using carrier-sense multiple accesses with collision avoidance (CSMA/CA), to transmit beacon frames for synchronization, and to provide reliable transmission.

NETWORK LAYER

The network layer is located between the MAC layer and Application Support sublayer. It provides the following functions:

- Starting a network
- Managing end devices joining or leaving a network
- Route discovery
- Neighbor discovery

APPLICATION SUPPORT SUBLAYER (APS)

The application support sublayer (APS) provides the services necessary for application objects (endpoints) and the ZigBee device object (ZDO) to interface with the network layer for data and management services. Some of the services provided by the APS to the application objects for data transfer are request, confirm, and response. Furthermore, the APS provides communication for applications by defining a unified communication structure (for example, a profile, cluster, or endpoint).

Application Object (Endpoint): An application object defines input and output to the APS. For example, a switch that controls a light is the input from the application object, and the output is the light bulb condition. Each node can have 240 separate application objects. An application object may also be referred to as an endpoint (EP).

ZigBee Device Object (ZDO): A ZigBee device object performs control and management of application objects. The ZDO performs the overall device management tasks:

- Determines the type of device in a network (for example, end device, router, or coordinator)
- Initializes the APS, Network Layer and Security Service Provider.
- Performs device and Service Discovery.
- Initializes Coordinator for Establishing a Network.
- Security Management.
- Network Management.
- Binding Management.

End node: Each end node or end device can have multiple EPs. Each EP contains an application profile, such as home automation, and can be used to control multiple devices or a single device. More to the point, each EP defines the communication functions within a device.

ZigBee addressing mode: ZigBee uses direct, group, and broadcast addressing for transmission of information. In direct addressing, two devices communicate directly with each other. This requires that the source device has both the address and endpoint of the destination device. Group addressing requires that the application assign a group membership to one or more devices. A packet is then transmitted to the group address in which the destination device lies. The broadcast address is used to send a packet to all devices in the network.

AT COMMANDS & ZigBee MODULE

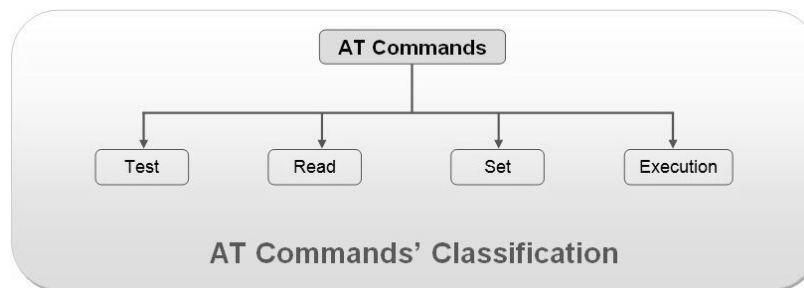
What are AT commands?

AT commands are commands which are used to control the modems where AT stands for Attention. These commands were derived from Hayes commands which were used by the Hayes smart modems. Every wireless as well as the dial up modems requires an AT command to interact with a computer machine. These AT commands along with other extended commands also require Hayes command set as a subset.

The Hayes command set is a specific command language originally developed for the Hayes Smart modem 300 baud modem in 1981. The command set consists of a series of short text strings which combine together to produce complete commands for operations such as dialing, hanging up, and changing the parameters of the connection. The vast majority of dialup modems use the Hayes command set in numerous variations. The command set covered only those operations supported by the earliest 300 bit/s modems. When new commands were required to control additional functionality in higher speed modems, a variety of one-off standards emerged from each of the major vendors.

Types of AT Command

There are 4 basic types of AT commands:



- **TEST:** The test command is utilized to check the compatibility of a command by a modem.
SYNTAX: *AT<command name>=?*
- **READ:** Read command is used for extracting the mobile or modem settings required for operations.
SYNTAX: *AT<command name>?*
- **SET:** This command is used to make changes into mobile phone or modem settings required for the operation.
SYNTAX: *AT<command name>=value1, value2, ..., value N*
- **EXECUTION:** As the name suggests, this command is used to execute the said operation.
SYNTAX: *AT<command name>=parameter1, parameter2, ..., parameter N*

ZigBee MODULE

In this ZigBee Trainer Kit we are using ProBee-ZE10 ZigBee module. The ProBee-ZE10 is an all-in-one ZigBee OEM module with integrated ZigBee core and radio/antenna circuits

together with high level software library. Using the ZE10, OEM manufacturers can easily and cost effectively integrates ZigBee functionality into target products in timely manner. The ZE10 OEM module is designed to meet specific requirements of low cost and low power wireless applications such as Home Automation, Smart Energy, Health care, Building Automation and Telecom Services by adopting the ZigBee technology. The ZE10 is based on the Ember EM250 ZigBee core and certified ZigBee Alliance based on ZigBee 2007 and ZigBee Pro stack.

Despite of its low cost and low power consumptions, the ZE10 provides high performance and robust data transfer capability for wide range of applications. The ZE10 can transfer data at up to 250 kbps and reach up to 1.6 km outdoor line-of-sight.

KEY FEATURES

- Integrated 2.4GHz, IEEE 802.15.4-compliant transceiver.
- ZigBee 2007/ ZigBee Pro stack.
- ZigBee core: Ember EM250.
- Transmit Power: +20dBm E.I.R.P (max.).
- Receiver Sensitivity: -102dBm @1% BER.
- Supply Voltage: 2.7~3.6 VDC.
- TX Current: 190mA@3.3V (max.).
- RX Current: 45mA@3.3V (max.).
- Working Distance: 1.6 km (1 mile).
- Various Antenna/Connector Options: Dipole 1/3/5dBi, U.FL, **Chip**.
- UART Signals Support: UART_TXD/RXD, RTS/CTS, DTR/DSR.
- 4 Analog Inputs.
- 13 Digital Inputs/Outputs.

ProBee-ZE10 - AT COMMAND REFERENCE

AT-Commands are classified into different sets according to their user. They are

- Standard AT Commands.
- Node Specific Information.
- Network Specific Parameters.
- Network Formation and Join.
- End-device.
- Data Transmission and Reception.
- Security.
- UART.
- GPIO.
- Firmware Upload and Help.

STANDARD AT COMMANDS

Command	AT
Description	Check the state of the connection between host and ProBee device.
Execute	AT<CR>
Response	OK<CR>

Command	ATB
Description	Enter the broadcast data mode. Every data is packetized at intervals of inter character timeout (S12) and transmit range is determined by broadcast range (S35).
Execute	ATB<CR>
Response	OK<CR>

Command	ATD
Description	Enter the Unicast data mode to specific node. If there's no argument after the command, it will connect to the preconfigured destination IEEE address (AT+DESTLA). Every data is packetized at intervals of inter character timeout (S12).
Execute	ATD<CR> ATD<NODEID><CR> ATD<IEEE><CR>
Response	OK<CR>

Command	ATE
Description	Echo (1) or No Echo (0).
Execute	ATE<number><CR>
Response	OK<CR>

Command	ATM
Description	Enter the multicast mode. If there's no argument after the command, it will connect to the preconfigured destination group ID (AT+DESTGROUPID). Every data is packetized at intervals of inter character timeout (S12).
Execute	ATM<CR> ATM<GROUPID><CR>
Response	OK<CR>

Command	ATO
Description	Return to data mode.
Execute	ATO<CR>
Response	

Command	ATR
Description	Response (1) or No Response (0).
Execute	ATR<number><CR>
Response	OK<CR>

Command	ATS
Description	Get/Set the S-registers.
Execute	ATS<CR> ATS<number>?<CR> ATS<number>=<value><CR>
Response	List of the <value><CR>OK <CR>OK<CR>/ERROR<CR> S-register and description

Command	ATZ
Description	Reset the node.
Execute	ATZ<CR>
Response	OK<CR>

Command	AT&F
Description	Factory reset.
Execute	AT&F<CR>
Response	OK<CR>

Command	+++
Description	Interrupt data mode. “+++” must be entered within 500ms.
Execute	+++
Response	OK<CR>

NODE SPECIFIC INFORMATION

Command	AT+LONGADDR or AT+LA
Description	Get the 64-bit IEEE address. (EUI address, long address) The IEEE address is a unique device address assigned during manufacturing. This address is unique to each physical device.
Execute	AT+LONGADDR?<CR>
Response	<IEEE address><CR> OK<CR>

Command	AT+SHORTADDR or AT+SA
Description	Get the 16-bit Node ID. (network address, short address) The node ID of “0000” is reserved for the coordinator. All other devices receive a randomly generated address from the coordinator or router device that allows the join. This value can change if an address conflict is detected where two devices are found to have the same address or a device leaves the network and later joins.
Execute	AT+SHORTADDR?<CR>
Response	<NODE ID><CR> OK<CR>

Command	AT+ADDRDISCOVERY or AT+AD
Description	Get the corresponding address of a node whose long or short address is known.

Execute	AT+ADDRDISCOVERY=<IEEE><CR>	AT+ADDRDISCOVERY=<NODE><CR>
Response	<NODE ID><CR> OK<CR>	<IEEE address><CR> OK<CR>

Command	AT+OPCH or AT+OC
Description	Get the operating channel number used for transmitting and receiving between RF modules. Uses 802.15.4 channel numbers. A value of "0" means the device has not joined a PAN and is not operating on any channel.
Execute	AT+OPCH?<CR>
Response	<VALUE><CR> OK<CR>

Command	AT+OPPANID or AT+OI
Description	Get the operating 16-bit PAN ID. If PAN ID is set specific value not 0s, operating PAN ID will equal designated PAN ID. A value of "FFFF" means the device is not joined a network.
Execute	AT+OPPANID?<CR>
Response	<VALUE><CR> OK<CR>

Command	AT+OPEPID or AT+OE
Description	Get the operating 64-bit extended PAN ID. A value of "FFFFFFFFFFFFFFFF" means the device is not joined a network.
Execute	AT+OPEPID?<CR>
Response	<VALUE><CR> OK<CR>

Command	AT+VERSION or AT+VR
Description	Get firmware version of the module. The firmware version returns a form of "PTVX.X".
Execute	AT+VERSION?<CR>
Response	<VALUE><CR> OK<CR>

Command	AT+PRODUCTNAME or AT+PN
Description	Get the product name. It can be used to distinguish among different hardware platforms.
Execute	AT+PRODUCTNAME?<CR>
Response	<VALUE><CR> OK<CR>

NETWORK SPECIFIC PARAMETERS

Command	AT+CHMASK or AT+CM	
Description	Get/Set the channel mask to use. Bits set as 1 indicate that this particular channel can be used. Bits set as 0 indicate that this particular channel should not be used. For example, a channel mask value of 0x00000001 indicates that only channel 0 should be used. Valid channels range from 12 to 25 inclusive. (14 channels are available.) This translates to a channel mask value of "0x03FFF000".	
Execute	AT+CHMASK?<CR>	AT+CHMASK=<value><CR>
Response	<VALUE><CR> OK<CR>	OK<CR> or ERROR<CR>
Default	03FFF000	
Range	00001000 - 03FFF000	

Command	AT+PANID or AT+PI	
Description	Get/Set the 16-bit PAN ID. If it is set to 0s at a coordinator, the coordinator will select a random 16-bit hexadecimal number as PAN ID. If it is set to 0s at a router/end device the router/end-device will try to join any existing network regardless of PAN ID.	
Execute	AT+PANID?<CR>	AT+PANID=<value><CR>
Response	<VALUE><CR> OK<CR>	OK<CR> or ERROR<CR>
Default	0000	
Range	0000 – FFFF	

Command	AT+EPID or AT+EI	
Description	Get/Set the 64-bit extended PAN ID. If the node is a coordinator and the EPID is set to 0s, it will create new extended PAN ID using its 64-bit IEEE address as extended PAN ID. If the node is a router or end-device and the EPID is set to 0x, it will try to join any existing network regardless of extended PAN ID.	
Execute	AT+EPID?<CR>	AT+EPID=<value><CR>
Response	<VALUE><CR> OK<CR>	OK<CR> or ERROR<CR>
Default	0000000000000000	
Range	0000000000000000 - FFFFFFFFFFFFFFFF	

Command	AT+GROUPID or AT+GI	
Description	Get/Set the 16-bit Group ID. It is used for multicast data reception.	
Execute	AT+GROUPID?<CR>	AT+GROUPID=<value><CR>
Response	<VALUE><CR> OK<CR>	OK<CR> or ERROR<CR>
Default	0000	
Range	0000 – FFFF	

Command	AT+POWER or AT+PW	
Description	Get/Set the TX power at which the RF module transmits conducted power.	
Execute	AT+POWER?<CR>	AT+POWER=<value><CR>
Response	<VALUE><CR> OK<CR>	OK<CR> or ERROR<CR>
Default	3	
Range	-43 – 3	

Command	AT+STACK or AT+ZS	
Description	Get/Set the ZigBee stack profile value. This value must be set the same on all devices that should join the same network. (0: Network specific, 1: ZigBee, 2: ZigBee Pro)	
Execute	AT+STACK?<CR>	AT+STACK=<value><CR>
Response	<VALUE><CR> OK<CR>	OK<CR> or ERROR<CR>
Default	2	
Range	0 - 2	

Command	AT+NODENAME or AT+NN	
Description	Get/Set the node name. It can be used to make a clear distinction role or functionality of local node from others.	
Execute	AT+NODENAME?<CR>	AT+NODENAME=<value><CR>
Response	<VALUE><CR> OK<CR>	OK<CR> or ERROR<CR>
Default	ProBee-ZE	
Range	Up to 16 characters	

NETWORK FORMATION AND JOIN

Command	AT+ASCAN or AT+AS	
Description	Perform a scanning for existing network. It can take up to 5 seconds. (CAUTION: It is possible to cause a link error if a node is a parent of an end-device.)	
Execute	AT+ASCAN<CR>	AT+ASCAN=<channel mask><CR>
Response	< CHANNEL> <PAN ID> <EPID> <joinable> <stack profile> List of the existing networks ... OK<CR>	

Command	AT+ESCAN or AT+ES	
Description	Perform a scanning for energy level of all channels. It can be used to avoid interference. It can take up to 5 seconds. The RSSI energy level ranges from approximately -100 to +127 in dBm at PTv1.4 or later. It is the real RSSI value plus an offset of +154 at PTv1.3 or earlier. (CAUTION: It is possible to cause a link error if a node is a parent of an end-device.)	

Execute	AT+ESCAN<CR> AT+ESCAN=<channel mask><CR>
Response	<CHANNEL> <RSSI > List of the RSSI for each channel ... OK<CR>

Command	AT+DSCAN or AT+DS
Description	Perform a scanning for existing nodes in the current network. A coordinator or router can have children nodes and they are presented by “CLDn”. It can be possible to place some conditions such as scan duration, node type and node name in order. The information of local node is presented by “*”. <i>(CAUTION: A ProBee-Z*10 with PTv1.2 or earlier cannot receive the response of AT+DSCAN from any ProBee-Z*10 with PTv1.3 or later. It cannot receive from any ProBee-Z*20, either. In this case, it is recommended to upgrade PTv1.2 to the latest version.)</i>
Execute	AT+DSCAN<CR> or AT+DSCAN=<scan duration><CR> or AT+DSCAN=<scan duration>,<node type><CR> or AT+DSCAN=<scan duration>,<node type>,<node name><CR>
Response	<node type> <IEEE address> <node ID> <version> <product name> <node name> <i>List of the existing nodes and information...</i> OK<CR>
Range	Scan duration: 1 – 10 Node type: 0 – 4 (Zero indicates all node types.)

Command	AT+RSSI or AT+RS
Description	Get RSSI and LQI information respectively for the delivery of the recent packet across the last hop of the incoming route. The LQI value ranges from 0 to 255 with the maximum value representing the best possible link quality. The RSSI value ranges from approximately -100 to +127 with each value representing the energy level in dBm at the radio's receiver. The sender of the recent packet and the parent are also displayed.
Execute	AT+RSSI?<CR>
Response	Sender: <node ID>, Parent: FFFF, RSSI: <RSSI value>, LQI: <LQI value><CR>OK<CR>

Command	AT+NODETYPE or AT+NT	
Description	Get/Set the node type. (0: none, 1: coordinator, 2: router, 3: end-device, 4: sleepy end device)	
Execute	AT+NODETYPE?<CR>	AT+NODETYPE=<value><CR>
Response	<VALUE><CR> OK<CR>	OK<CR> or ERROR<CR>
Default	0	
Range	0 - 4	

Command	AT+PERMIT or AT+PJ	
Description	Permit other nodes to join the network. (0~254s, 255=continuance) An end-device cannot perform it	
Execute	AT+PERMIT<CR>	AT+PERMIT=<value><CR>
Response	<VALUE><CR> OK<CR>	OK<CR> or ERROR<CR>
Range	0 - 255	

Command	AT+LEAVE or AT+LV	
Description	Leave the current network	
Execute	AT+LEAVE<CR>	
Response	OK<CR>	

END-DEVICE

Command	AT+SLEEP or AT+SM	
Description	Enter the sleep mode immediately if the node is a sleepy end-device.	
Execute	AT+SLEEP<CR>	
Response	OK<CR>	

Command	AT+PARENTLA or AT+PL	
Description	Get the 64-bit IEEE address of the parent node.	
Execute	AT+PARENTLA?<CR>	
Response	<IEEE address><CR> OK<CR>	

Command	AT+PARENTSA or AT+PS	
Description	Get the 16-bit Node ID of the parent node.	
Execute	AT+PARENTSA?<CR>	
Response	<node ID><CR> OK<CR>	

Command	AT+CHILDTABLE or AT+CT	
Description	Get the child table composed of the end-devices.	
Execute	AT+CHILDTABLE?<CR>	
Response	<node type> <IEEE address> <node ID> <i>List of the child</i>	

DATA TRANSMISSION AND RECEPTION

Command	AT+DESTLA or AT+DL	
Description	Get/Set the 64-bit destination IEEE address for Unicast data transmission.	
Execute	AT+DESTLA?<CR>	AT+DESTLA=<value><CR>

Response	<value><CR> OK<CR>	OK<CR> or ERROR<CR>
Default	0000000000000000	
Range	0000000000000000 - FFFFFFFF	

Command	AT+DESTGROUPID or AT+DG	
Description	Get/Set the 16-bit destination Group ID for multicast data transmission.	
Execute	AT+DESTGROUPID?<CR> AT+ DESTGROUPID=<value><CR>	
Response	<value><CR> OK<CR>	OK<CR> or ERROR<CR>
Default	0000	
Range	0000 - FFFF	

Command	AT+SINKINFO or AT+SI	
Description	Get the long and short addresses of sink node. If there is no sink node in the network, the addresses will be "FFFFFFFF" and "FFFF".	
Execute	AT+SINKINFO?<CR>	
Response	<IEEE address>,<nodeID><CR> OK<CR>	

Command	AT+MAXPAYLOAD or AT+MP	
Description	Get the maximum size of the payload that the Application Support sub-layer will accept. The size depends on the security level in use.	
Execute	AT+MAXPAYLOAD?<CR>	
Response	<value><CR> OK<CR>	

Command	AT+TRASNSMITMODE or AT+TM	
Description	Get/Set the data transmission mode. This mode is maintained after rebooting. (0:none, 1: Unicast to destination long address, 2: multicast to destination group ID, 3:broadcast to all nodes, 4: Unicast to sink node)	
Execute	AT+TRANSMITMODE?<CR> AT+TRANSMITMODE=<value><CR>	
Response	<value><CR> OK<CR>	OK<CR> or ERROR<CR>
Default	0	
range	0 – 4	

Command	AT+UNICAST or AT+UC	
Description	Send Unicast message to specific node using long or short address of remote node. Up to 90 bytes (or 72 bytes when security enabled) are sent to the node at once.	
Execute	AT+UNICAST=<IEEE>,<data><CR> AT+UNICAST=<nodeID>,<data><CR>	

Response	OK<CR>
-----------------	--------

Command	AT+MULTICAST or AT+MC
Description	Send multicast message to specific group. Every node belongs to the group after being booted up if it is designated to a group ID (AT+GROUPID).
Execute	AT+MULTICAST=<groupID>,<data><CR>
Response	OK<CR>

Command	AT+BROADCAST or AT+BC
Description	Send broadcast message to the all nodes joining the network. Transmission range is determined by broadcast range (S35).
Execute	AT+BROADCAST=<data><CR>
Response	OK<CR>

Command	AT+REMOTE or AT+RC
Description	Get/Set the configuration values of remote node using long or short address. AT+REMOTE<CR> shows the list of available commands.
Execute	AT+REMOTE=<addr>,<cmd>?<CR> AT+REMOTE=<addr>,<cmd>=<value><CR>
Response	<value><CR> OK<CR> OK<CR>

SECURITY

Command	AT+SECURITY or AT+SE
Description	Get/Set the security status. (0: turning security off, 1: on) Every node must be under the same security level in the network.
Execute	AT+SECURITY?<CR> AT+SECURITY=<value><CR>
Response	<value><CR> OK<CR> or ERROR<CR> OK<CR>
Default	0
Range	0 - 1

Command	AT+LINKKEY or AT+LK
Description	Get/Set the 128-bit link key. (0 or 0s: generate 128-bit random number).
Execute	AT+LINKKEY?<CR> AT+LINKKEY=<value><CR>
Response	<value><CR> OK<CR> or ERROR<CR> OK<CR>
Default	FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF

Command	AT+LINKKEY or AT+LK
Description	Get/Set the 128-bit link key. (0 or 0s: generate 128-bit random number).

Execute	AT+LINKKEY?<CR>	AT+LINKKEY=<value><CR>
Response	<value><CR> OK<CR>	OK<CR> or ERROR<CR>
Default	FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF	

Command	AT+NWKEY or AT+NK	
Description	Get/Set the 128-bit network key. (0 or 0s: generate random 128-bit number).	
Execute	AT+NWKEY?<CR>	AT+NWKEY=<value><CR>
Response	<value><CR> OK<CR>	OK<CR> or ERROR<CR>
Default	FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF	

Command	AT+UPDATEKEY or AT+UK	
Description	Update the network key with a new generated key. (possible only for coordinator) It is applied after 30s later because sending too quickly may cause a sleepy end-device to miss the alternate encryption key and only get the switch key message, which means it will be unable to change to the new network key.	
Execute	AT+UPDATEKEY<CR>	
Response	OK<CR> or ERROR<CR>	

UART

Command	AT+BAUDRATE or AT+UB	
Description	Get/Set the serial port speed for communication between the module serial port and host.	
Execute	AT+BAUDRATE?<CR>	AT+BAUDRATE=<value><CR>
Response	<value><CR> OK<CR>	OK<CR> or ERROR<CR>
Default	9600	
Range	300 - 230400	

Command	AT+DATABIT or AT+UD	
Description	Get/Set the serial data bit setting on the module.	
Execute	AT+DATABIT?<CR>	AT+DATABIT=<value><CR>
Response	<value><CR> OK<CR>	OK<CR> or ERROR<CR>
Default	8	
Range	7 - 8	

Command	AT+PARITY or AT+UP	
Description	Get/Set the serial parity setting on the module.	
Execute	AT+PARITY?<CR>	AT+PARITY=<value><CR>
Response	<value><CR> OK<CR>	OK<CR> or ERROR<CR>

Default	0
Range	0 - 2

Command	AT+STOPBIT or AT+US	
Description	Get/Set the serial stop bit setting on the module.	
Execute	AT+STOPBIT?<CR>	AT+STOPBIT=<value><CR>
Response	<value><CR> OK<CR>	OK<CR> or ERROR<CR>
Default	1	
Range	1 - 2	

Command	AT+FLOWCTRL or AT+UF	
Description	Get/Set flow control status. (0: none, 1: software, 2: hardware flow control)	
Execute	AT+FLOWCTRL?<CR>	AT+FLOWCTRL=<value><CR>
Response	<value><CR> OK<CR>	OK<CR> or ERROR<CR>
Default	0	
Range	0 - 2	

GPIO

Command	AT+GPIO or AT+IO	
Description	Get/Set the configuration status of GPIO individually or all at once. Only GPIO3-6 can be set 4, others are available 0-3 and 5.	
Execute	AT+GPIO?<CR> or AT+GPIO<number>?<CR>	AT+GPIO=<values><CR> or AT+GPIO<number>=<value><CR>
Response	<value><CR> OK<CR>	OK<CR> or ERROR<CR>
Default	5550000055550	
Range	GPIO3-6: 0-5, the others: 0-3 and 5 0 Disabled (unmonitored digital input) 1 Digital input, monitored 2 Digital output, default low 3 Digital output, default high 4 Analog input, single ended (only valid for GPIO3~6) 5 Reserved for pin-specific alternate functionalitie.	

Command	AT+DIO	
Description	Get/Set the digital IO value individually or all at once. All GPIO pins can be used as digital IO and it is only possible to set 0 or 1. If there are some pins used analog input, they are displayed by “*”.	
Execute	AT+DIO?<CR> or AT+DIO<number>?<CR>	AT+DIO=<values><CR> or AT+DIO<number>=<value><CR>
Response	<value><CR> OK<CR>	OK<CR> or ERROR<CR>
Default	0000000000000	

Range	0000000000000 - 111111111111	
Command	AT+AI	
Description	Get the analog input value individually or all at once. GPIO3~6 are possible to set ADC and it's read only. If there are some pins used digital IO, they are displayed by "****".	
Execute	AT+AI?<CR>	AT+AI<number>?<CR>
Response	<value>,<value>,<value>,<value><CR> OK<CR>	<value><CR> OK<CR>
Command	AT+ SOURCELA or AT+SL	
Description	Get/Set the 64-bit source IEEE address for GPIO remote control. It is possible to limit on the implementation of this functionality by setting specific address. If set to default value, "FFFFFFFFFFFFFFFF", the node allows that any node can control the GPIO status.	
Execute	AT+SOURCELA?<CR>	AT+SOURCELA=<value><CR>
Response	<value><CR> OK<CR>	OK<CR> or ERROR<CR>
Default	FFFFFFFFFFFFFFFF	
Range	0000000000000000 - FFFFFFFFFFFFFFFFFF	

FIRMWARE UPLOAD AND HELP

Command	AT+BOOTLOAD or AT+BL	
Description	Start Boot loader to upload firmware. (blank: local, 1: passthru, 2: clone).The device leaves the AT command line and enters the Boot loader menu for uploading new ProBee firmware. There's no restrict to use local upload but a sleepy end-device cannot upload to remote node over the air. The current UART should be set 115200bps, 8 data bits, no parity, 1 stop bit and none flow control.	
Execute	AT+BOOTLOAD<CR>	AT+BOOTLOAD=<number>,<IEEE><CR>
Response	<i>Bootloader menu</i> image ...<CR> or ... Serial upload complete	Please start .ebl upload <i>Bootloader menu</i> ... Start cloning ...<CR> Bootload Complete.
Command	AT+HELP or AT+HP	
Description	Description Show all the available commands	
Execute	AT+HELP<CR>	
Response	Available AT Commands ... OK<CR>	

S-REGISTERS

DISPLAY OPTIONS

Register	S11
Description	Get/Set the display mode. It determines to display incoming message in command mode.
Execute	ATS11?<CR> ATS11=<value><CR>
Response	<value><CR> OK<CR>
Default	0
Range	0 - 1

Register	S12
Description	Get/Set the command echo mode.
Execute	ATS12?<CR> ATS12=<value><CR>
Response	<value><CR> OK<CR>
Default	0
Range	0 - 1

Register	S13
Description	Get/Set the command responses mode.
Execute	ATS13?<CR> ATS13=<value><CR>
Response	<value><CR> OK<CR>
Default	0
Range	0 - 1

Register	S14
Description	Get/Set the LED mode. It causes the LED operation if the device is ZS10 or ZE10 set GPIO1, 2 to specific functionality. 0: LEDs are always OFF regardless of configuration or status. 1: LEDs are OFF when the node is under sleep status. 2: LEDs always operate by the case.
Execute	ATS14?<CR> ATS14=<value><CR>
Response	<value><CR> OK<CR>
Default	0
Range	0 - 2

NETWORK SETTINGS

Register	S21
Description	Get/Set the inter character timeout in milliseconds. It determines to send data packet after this time.

Execute	ATS21?<CR>	ATS21=<value><CR>
Response	<value><CR> OK<CR>	OK<CR>
Default	100	
Range	0 - 10000	

Register	S22	
Description	Get/Set the timeout to permit joining. It determines the time in seconds that the device will allow other devices to join to it when it starts network but an end-device cannot perform it. If set to 255, the device will always permit joining.	
Execute	ATS22?<CR>	ATS22=<value><CR>
Response	<value><CR> OK<CR>	OK<CR>
Default	255	
Range	0 - 255	

Register	S23	
Description	Get/Set the power boost mode of the module. If enabled, boost mode improves sensitivity by 1dB and increases output power by 2dB, improving the link margin and range.	
Execute	ATS23?<CR>	ATS23=<value><CR>
Response	<value><CR> OK<CR>	OK<CR>
Default	0	
Range	0 - 1	

DATA TRANSMISSION

Register	S31	
Description	Get/Set the application-ACK mode. If it is enabled, a sender node retransmits data packet when it fails to receive ACK from receiver node. This value must be set same on the source and destination node for data transmission.	
Execute	ATS31?<CR>	ATS31=<value><CR>
Response	<value><CR> OK<CR>	OK<CR>
Default	0	
Range	0 - 1	

Register	S32	
Description	Get/Set the transmission retries. A sender tries to retransmit up to this value when transmission fails. (CAUTION: It is recommended to set the smallest value that the application can tolerate in broadcast/multicast data transmission.)	
Execute	ATS32?<CR>	ATS32=<value><CR>
Response	<value><CR>	OK<CR>

	OK<CR>
Default	5
Range	0 - 99

Register	S33
Description	Get/Set the transmission timeout in milliseconds. A sender retransmits data packet after this time when transmission fails. (CAUTION: It is recommended to set this value more than 1000 in broadcast/multicast data transmission.)
Execute	ATS33?<CR> ATS33=<value><CR>
Response	<value><CR> OK<CR>
Default	500
Range	0 - 10000

Register	S34
Description	Get/Set the indirect transmission timeout in milliseconds. It is the maximum amount of time that the MAC will hold a message for indirect transmission to a child.
Execute	ATS34?<CR> ATS34=<value><CR>
Response	<value><CR> OK<CR>
Default	3000
Range	0 - 30000

Register	S35
Description	Get/Set the broadcast range. ZigBee specifies three different broadcast addresses that reach different collections of nodes. Broadcasts are normally sent only to routers. Broadcasts can also be forwarded to end-devices, either all of them or only those that do not sleep. Broadcasting to sleepy end-devices is both significantly more resource intensive and significantly less reliable than broadcasting to routers. 0: send to all routers 1: send to all non-sleepy nodes 2: send to all nodes in the network
Execute	ATS35?<CR>ATS35=<value><CR>
Response	<value><CR>OK<CR>
Default	2
Range	0 - 2

Register	S36
Description	Get/Set the maximum number of hops the message will be relayed. The message will be delivered to all nodes within this number of hops of the sender. A value of zero is converted to maximum value (30). It is used to multicast/broadcast.

Execute	ATS36?<CR>	ATS36=<value><CR>
Response	<value><CR> OK<CR>	OK<CR>
Default	0	
Range	0 - 10	

Register	S37	
Description	Get/Set the number of hops that the message will be forwarded by devices that are not members of the group. A value of zero, 7 or greater is treated as infinite	
Execute	ATS37?<CR>	ATS37=<value><CR>
Response	<value><CR> OK<CR>	OK<CR>
Default	0	
Range	0 - 7	

Register	S38	
Description	Get/Set the receive message mode. If it is enabled, a local node will receive multicast or broadcast messages from itself.	
Execute	ATS38?<CR>	ATS38=<value><CR>
Response	<value><CR> OK<CR>	OK<CR>
Default	0	
Range	0 - 1	

Register	S39	
Description	Get/Set the receive mode. If S39 is enabled, a local node will only receive the same type of messages with its transmission mode. For example, if local node's transmission mode is broadcast mode (AT+TM=3), it will receive only broadcast messages and won't receive unicast or multicast messages.	
Execute	ATS39?<CR>	ATS39 =<value><CR>
Response	<value><CR> OK<CR>	OK<CR>
Default	0	
Range	0 - 1	

GPIO

Register	S41	
Description	Get/Set the GPIO pull up/down option. If GPIOs are set to DI (Digital Input), it is possible to pull-up or pull-down with this value. Pull-down = 0, Pull-up = 1.	
Execute	ATS41?<CR>	ATS41=<value><CR>
Response	<value><CR> OK<CR>	OK<CR>

Default	0
Range	0 - 1

Register	S42
Description	Get/Set the GPIO monitoring period in seconds. If it is enabled, a node sends GPIO values to a remote node based on preconfigured destination IEEE address periodically and it is possible to monitor the GPIO status at the remote node. A remote node can also be the sink node according to GPIO monitoring node (S43). If set to 0, GPIO monitoring is disabled.
Execute	ATS42?<CR> ATS42=<value><CR>
Response	<value><CR> OK<CR>
Default	0
Range	0 - 10000

Register	S43
Description	Get/Set the GPIO monitoring node. If S43 is 0, the monitoring node is selected by AT+DESTLA or AT+DL command. If S43 is 1, the sink node is set as the monitoring node
Execute	ATS43?<CR> ATS43=<value><CR>
Response	<value><CR> OK<CR>
Default	0
Range	0 - 10000

Register	S44
Description	Get/Set the GPIO change detection sampling mode. If it is enabled, a node sends GPIO values to a remote node based on preconfigured destination IEEE address immediately when the status of GPIO changes. A remote node can also be the sink node according to GPIO monitoring node (S43). If set to 0, GPIO change detection sampling is disabled.
Execute	ATS44?<CR> ATS44=<value><CR>
Response	<value><CR> OK<CR>
Default	0
Range	0 - 1

Register	S45
Description	Get/Set the GPIO remote control mode. If any GPIO pin is set to digital input, a node can control the I/O status of a remote node if the I/O pin of the remote node is set to digital output. It is possible to use when GPIO monitoring is enabled (GPIO monitoring period, S42 > 0) or GPIO change detection sampling is enabled.(GPIO change detection sampling mode, S44 = 1)
Execute	ATS45?<CR> ATS45=<value><CR>+

Response	<value><CR> OK<CR>
Default	0
Range	0 - 1

END DEVICE

Register	S51
Description	Get/Set the poll timeout for a parent node in seconds. If an end-device or a sleepy end-device does not send a poll request to its parent coordinator or router within the poll timeout, the end-device is removed from the child table.
Execute	ATS51?<CR>ATS51=<value><CR>
Response	<value><CR>OK<CR>
Default	60
Range	0 – 4177920

Register	S52
Description	Get/Set the sleepy end-device poll period in milliseconds. Adaptive polling may allow the sleepy end-device to poll more rapidly for a short time when receiving RF data.
Execute	ATS52?<CR> ATS52=<value><CR>
Response	<value><CR> OK<CR>
Default	100
Range	0 - 10000

Register	S53
Description	Get/Set the sleep duration in quarter seconds. An end-device sleeps for this time and turns off radio every sleep cycle. If a node sleeps over poll timeout of parent node, the device will be removed from the child table.
Execute	ATS53?<CR>ATS53=<value><CR>
Response	<value><CR>OK<CR>
Default	8
Range	0 - 16711680

Register	S54
Description	Get/Set the standby timeout in seconds. When an end-device wakes up and polls with the parent node, the node will wait for another data from the ZigBee network or from the serial port for S54 before it goes to sleep mode again. If any serial or RF data is sent or received, the timer will reset and wait for next again.
Execute	ATS54?<CR> ATS54=<value><CR>
Response	<value><CR> OK<CR> OK<CR>

Default	5
Range	0 - 10000

Register	S55
Description	Get/Set the rejoin timeout. If it is enabled, an end-device tries to rejoin the current network when contact with the network has been lost for setting value in seconds. The most common usage case is when an end-device can no longer communicate with its parent and wishes to find a new one. A value of zero, don't try to rejoin.
Execute	ATS55?<CR> ATS55=<value><CR>
Response	<value><CR> OK<CR>
Default	10
Range	0 - 10000

Register	S56
Description	Get/Set the end-device poll period in seconds. Adaptive polling may allow the end device to poll more rapidly for a short time when receiving RF data.
Execute	ATS56?<CR> ATS56=<value><CR>
Response	<value><CR> OK<CR>
Default	5
Range	0 - 10000

SINK

Register	S61
Description	Get/Set the sink node and the sink advertise period in seconds. When a node is defined as a sink, it can multicast its address to the rest of the network in this period. If set to 0, sink advertise is disabled.
Execute	ATS61?<CR> ATS61=<value><CR>
Response	<value><CR> OK<CR>
Default	10
Range	0 - 10000

Register	S62
Description	Get/Set the period to send Many-to-One Route Request. Many-to-One Route Request (MTOR) creates routes from every node in the network back to this node. It is useful when this node acts as a sink node
Execute	ATS62?<CR> ATS62=<value><CR>
Response	<value><CR> OK<CR>
Default	0
Range	0 - 10000

ZIGBEE TRAINER HARDWARE AND SOFTWARE DETAILS

POWER SECTION

In Power Supply section, need to connect an external 5V-DC adaptor to source the power to power supply section. ZigBee trainer is working with 5V DC & 3.3V DC. In power supply section have 3.3V regulator to supply the regulated voltage to ZigBee Module and its peripherals. Five volt is used to USB to Serial converter Section, Relay Section and Buzzer Section and switch SW6 used to ON & OFF the power.

ZigBEE SECTION

The heart of this trainer is the ZigBEE section, it carrying ProBee ZE10C Module. ZigBee have 28 pin DIP package types all GPIO pins are rooted internal for specific peripherals. It operates at the voltage of 3.3V DC and UART is used to communicate to the ZigBee with computer or microcontroller. Configuration table is given below.

PIN NO:	ZIGBEE PIN	PERIPHERAL	PIN NO:	ZIGBEE PIN	PERIPHERAL
1	GND		15	GND	
2	NC		16	GND	
3	GPIO_1	ZIGBEE POWER LED	17	GPIO_12	BUZZER
4	GPIO_2	ZIGBEE STATUS LED	18	GPIO_11	RELAY
5	GPIO_0	LED1	19	GPIO_10	SWITCH-3
6	FACTORY RESET	FACTORY RESET SWITCH	20	UART_RXD	USB TO SERIAL TXD
7	GPIO_3/ADC_0	TEMPERATURE SENSOR	21	UART_TXD	USB TO SERIAL RXD
8	GPIO_4/ADC_1	LDR SENSOR	22	GPIO_8	SWITCH-2
9	H/W RESET	H/W RESET SWITCH	23	GPIO_9	SWITCH-1
10	GPIO_5/ADC_2	VARIABLE RESISTOR	24	GPIO_7	LED2
11	GPIO_6/ADC_3	LED3	25	NC	
12	+3.3V	POWER SUPPLY 3.3V	26	NC	
13	NC		27	NC	
14	GND		28	GND	

USB TO SERIAL CONVERTOR

Device that adds serial port to the computer it plugs into the USB port and contains one or more serial port sockets. It convert USB standard to TTL UART standard. Here we used FTDI driver and operated at +5V DC.

FACTORY RESET: To restore the ZE10 device settings to factory default, this input needs to be pressed for more than two seconds. It will be applied after the input released again. If

this input is pressed while the ZE10 starts, the local node enters boot loader menu and it is possible to upload the firmware.

RESET: To reset the module, all configurations remain same.

POWER LED: By default GPIO1 is set to Power Status of the module, this goes high when power is supplied. If the node is set to an end-device, it will blink periodically.

STATUS LED: By default GPIO5 is set to Status of the module, this goes high when the device joins a network, blinks when joining is permitted, and turns OFF when leaves the network.

PERIPHERALS SECTION

1. OUTPUT LED UNIT

In OUTPUT LED units have three LEDs that can accessible by the user. Initially all LEDs are active low state if you want to enable a particular LED then you need make the pin active high in ZigBee side.

LED NAME	ZigBee PIN	LED-ON	LED-OFF
LED1	PIN-05	ACTIVE HIGH	ACTIVE LOW
LED2	PIN-24	ACTIVE HIGH	ACTIVE LOW
LED3	PIN-11	ACTIVE HIGH	ACTIVE LOW

Active High = Logical One

Active Low = Logical Zero

2. INPUT SWITCH UNIT

In input switch units have three switches that can accessible by the user. Switches are the input device to give logical state (0 & 1) to the ZigBee module. Logical Zero represents active Low signal (switch-OFF) and Logical One represent Active High Signal (switch-ON).

SWITCH	ZigBee PIN	ON	OFF
SWITCH-1	PIN-23	ACTIVE HIGH	ACTIVE LOW
SWITCH-2	PIN-22	ACTIVE HIGH	ACTIVE LOW
SWITCH-3	PIN-19	ACTIVE HIGH	ACTIVE LOW

3. BUZZER UNIT

Buzzer is also onboard peripheral. This is also accessible by the user, if the user wants to turn on the buzzer then apply active high (logical one) signal to corresponding pin of the ZigBee Module.

BUZZER	ZigBee PIN	ON	OFF
	PIN-17	ACTIVE HIGH	ACTIVE LOW

4. RELAY UNIT

Relay is also an onboard peripheral. This is also accessible by user, if the user wants study the relay operation then the jumpers (J3, J4, and J5) should place pin 1 & pin 2. If the user want to connect an external device to the relay jumpers (J3, J4, and J5) should place pin 2 & pin 3.

RELAY	ZigBee PIN	ON	OFF
	PIN-18	ACTIVE HIGH	ACTIVE LOW

5. ADC UNIT

In ADC unit you can have a 10k variable port and this is connected to the analog to digital converter pin of ZigBEE module. User can vary the voltage from zero to 1.21V because the ZigBee module accepts 1.21V maximum.

ADC	VALUE	VOLTAGE	ZigBee PIN
PR1	10K	0 - 1.21V	PIN-10

6. LDR UNIT

LDR Light Dependence Resistor is an on board peripheral used to measure the light intensity. This is also accessible by user, if the user wants to use this device first enable the ADC pin of ZigBee. According to the light intensity, LDR varies its resistance and corresponding analog voltage (range 0 – 1.21V) to ADC pin.

LDR	VOLTAGE	ZigBee PIN
LDR1	0 - 1.21V	PIN-08

7. TEMPERATURE SENSOR

This is also a user accessible peripheral measure the temperature. This sensor is also connected to ADC of the ZigBee Modem and it can measure the temperature from 0°C to 120°C. This temperature sensor converts the temperature factor into analog voltage levels.

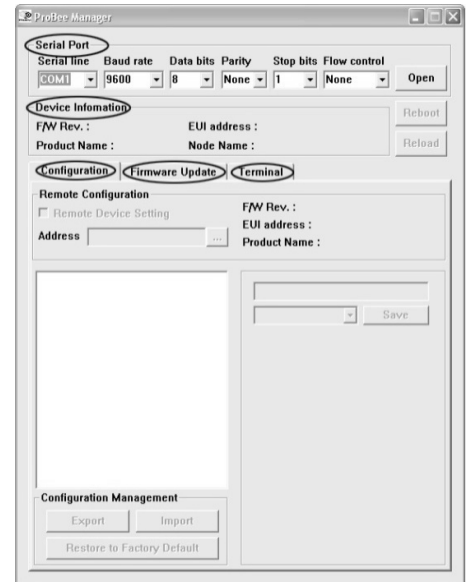
SENSOR	VALUE	VOLTAGE	ZigBee PIN
TEMP	0°C to 120°C	0 - 1.21V	PIN-07

For Configuring all peripherals of the ZigBee trainer AT+GPIO=2554442211122.

SOFTWARE DETAILS

PROBEE MANAGER FOR CONFIGURATION

This is a PC utility helps that a user can configure several setting values of local node and remote node easily without a terminal program. It is possible to export current configuration and import it to other nodes. It is also possible to restore to factory default configuration. It provides users with functions to upgrade firmware of not only local node but also remote node and clone the firmware of local node to remote node.



Serial Port: it is used to configure the serial port parameters and open serial port directly from ProBee Manager Utility.

Device Information: After communicate with ZigBee Modem. It shows the basic device information.

Configuration: The Configuration Pane appears when the 'Configuration' tab is selected. To change configuration, users need to find and modify the property and then apply changes. The properties are categorized according to the function. Selecting a category unfolds or folds the property list. The name, the current value and the description of the property are shown at the right of the configuration pane by selecting a property in the property list. It is possible to edit and save the current value of editable property. Whenever property is modified, it is necessary to save the change by clicking the 'Save' button in order not lose it moving to other properties. To apply changes, please reboot the device by clicking 'Reboot' button. The 'Save' button is disabled for read-only property.

Firmware Update: This window is used to update the Firmware of the ZigBee.

Terminal: This window is similar to hyper terminal page to send and receive data.

INSTALLATION OF ZIGBEE TRAINER HARD WARE SETUP AND DRIVER INSTILLATION

Note: Before you install ZigBee Trainer, Please make sure that your Adaptor output voltage is 5V-DC (more than 5V is not preferable).
Keep the ZigBee Trainer Power switch to OFF Position.

1. Connect the Adaptor to the ZigBee Trainer.
2. Connect the USB Cable to the ZigBee Trainer and PC.
3. Turn on the Adaptor.
4. Turn on the ZigBee Trainer kit.
5. First you need to install USB driver and SERIAL driver as shown below.

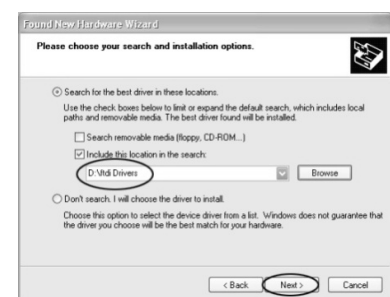
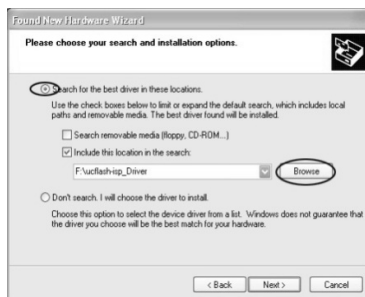
6. After turn on the ZigBee Trainer, a message window is appearing on the right corner of the monitor.



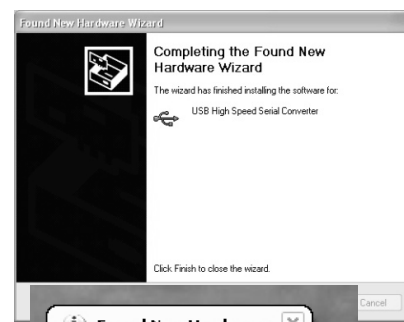
7. After a while New Hardware Instillation Wizard is appear on the screen and select the second option and apply Next button (USB driver instillation).



8. After applying the Next button, Driver installation window will appear. Select the first option and use the Browse button to locket the driver manually. After locating the driver (CD-ROM) and apply OK then apply Next button.



9. After applying the Next button Windows (OS) search driver automatically. After complete this process apply Finish button.

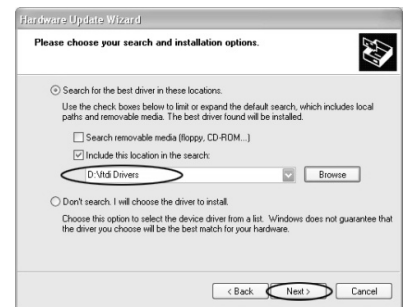


10. After applying the Finish button a new message window appear on the right corner of the monitor.

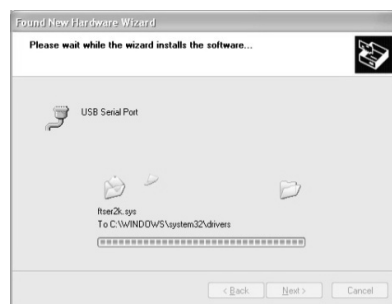


11. After a while New Hardware Installation Wizard is appear on the screen and select the second option and apply Next button (SERIAL driver installation).

12. After applying the Next button, Hardware Update Wizard will appear. Select the first option and use the Browse button to locket the driver manually. After locating the driver (same folder from CD-ROM) and apply OK then apply Next button.



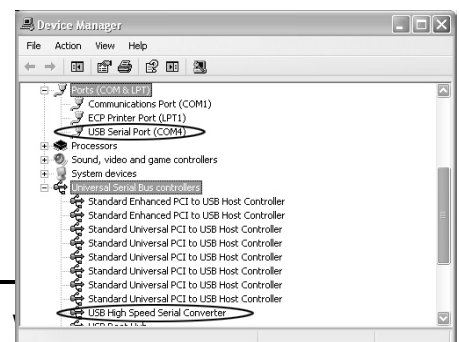
13. After applying the Next button Windows (OS) search driver automatically. After complete this process apply Finish button.



14. After applying the Finish button a new message window appear on the right corner of the monitor. It's showing the hardware is installed and ready to use.



15. Then go to the Device Manager and click the Universal Serial Bus Controller you can see USB High Speed Serial Converter Device then click the Port (Com & LPT) you can see USB Serial Port Device and Port Address also.

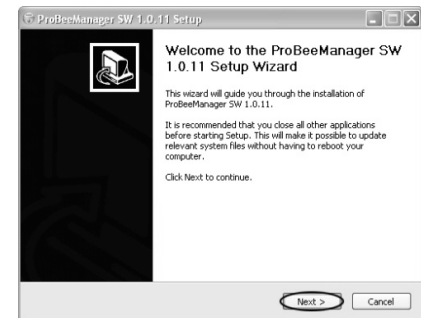


PROBEE MANAGER INSTILLATION

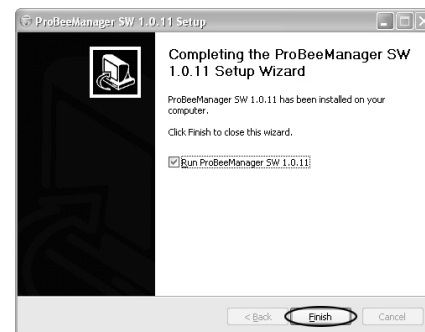
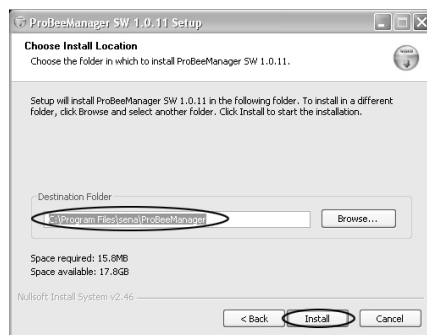
ProBee Manager is a PC utility helps that a user can configure several setting values of local node and remote node easily without a terminal program. It is possible to export current configuration and import it to other nodes. It is also possible to restore to factory default configuration. It provides users with functions to upgrade firmware of not only local node but also remote node and clone the firmware of local node to remote node.

1. Run ProBee Manager.exe.

2. After that ProBee Manager setup window will appear and apply Next button.



3. After applying the Next button a new window will appear. Here we can select the installation folder other ways keep as default and apply Install button. After finished the installation apply Finish Button



4. After installation completed the ProBee Manager pop up automatically. This software has five different user friendly menus.

a) **Serial Port**: used to configure the serial port parameters and open serial port directly.

b) **Device Information**: After communicate with ZigBee. It shows the basic information display here.

c) **Configuration**: This window is used to configure ZigBee without using AT-Command.

- d) **Firmware Update**: This window is used to update the Firmware of ZigBee.
- e) **Terminal**: This window is similar to hyper terminal page to send and receive data.



Experiment No.1

Experiment No.1

Name

Configuration of the ZigBee module

Objective

To study how to configure the ZigBee Module

1. ZigBee as Coordinator
2. ZigBee as Router
3. ZigBee as End device

Equipments

- ZigBee Trainer Kit
- USB Cable
- Power Supply

Procedure

CONFIGURATION THE ZIGBEE MODULE

A ZigBee Network consists of a coordinator, routers and end-devices. A minimal ZigBee network consists of one coordinator and multiple end-devices which directly connect to the coordinator. For larger ZigBee networks, routers are required to provide redundant routings to form mesh networks.

1. **ZIGBEE COORDINATOR:** The coordinator forms the root of the network tree and might bridge to other networks. There is exactly one ZigBee coordinator in each network since the coordinator starts the ZigBee network itself. It also stores information about the network, including acting as the Trust Center & repository for security keys.
2. **ZIGBEE ROUTER:** The routers are able to route messages between other devices. Since the coordinator provides the routing functionality too, the routers are optional to form a ZigBee network.

3. **ZIGBEE END-DEVICE:** The end-device contains just enough functionality to talk to the parent node which is either the coordinator or a router. It cannot relay data from other devices.

The ZE10 can be configured to work either as a coordinator, a router and end-device. Once the operation mode of the ZE10 is selected, corresponding parameters need to be set up appropriately. This section explains how to configure the ZE10 operation mode and other parameters related to the ZigBee network itself.

NODE TYPE	Recommendation when to use ZE10
COORDINATOR	It should be selected when it is first time to establish the network.
ROUTER	It is recommended when there is a network already and it needs to be expanded. A router is suitable for fast and stable data transmissions and receptions.
END-DEVICE	It is recommended when it doesn't need to expand the network anymore and there are enough nodes as parents. An end-device consumes less power than a router.
SLEEPY END-DEVICE	It is recommended when the data transmissions occur rarely and in very small quantities. A sleepy end-device can lose data if sleep duration is too long, but it is possible to consume extremely low power as low as 2uA.

Experiment No.1A

Name

Configuration of the ZigBee Module as a Coordinator

Objective

To study how to configure the ZigBee Module as a Coordinator

Equipments

ZigBee Trainer Kit

Procedure

CONFIGURING ZIGBEE AS COORDINATOR

A coordinator relays messages and can act as a parent to other nodes. Every personal area network (PAN) must be started by a node acting as the coordinator. In general, this device is always powered on. To set up the ZE10 to work as the coordinator, please follow the steps as below

- Configure the ZE10 node type as the Coordinator
- Optionally, select the channel mask. Otherwise, the ZE10 will use the default channel mask.
- Optionally, set up the PAN ID or the extended PAN ID. Otherwise, the ZE10 will generate the PAN ID or the extended PAN ID automatically.
- Permit joining when other router or end-device tries to join the ZigBee network.

A. NODE TYPE CONFIGURATION

The ZE10 module can be configured as a ZigBee coordinator by the AT command:

- $AT+NODETYPE = n$ or $AT+NT = n$
where n is 0 = none, 1 = coordinator, 2 = router, 3 = end-device,
4 = sleepy end-device.

After the operation mode is selected, ATZ command should be performed to reset the unit to have the change applied.

AT-COMMAND	DESCRIPTION
AT+NODETYPE=1<CR>	Set node type as the coordinator
OK<CR>	Response
ATZ<CR>	Apply the change
OK<CR>	Response

B. CHANNEL MASK CONFIGURATION

When the ZigBee coordinator starts, it scans predefined frequency channels to find the best frequency channel to use without causing interference with other wireless devices. The user is able to specify the channels to use with the AT command:

- *AT+CHMASK = n or AT+CM = n*
where *n* = channel mask to use. Default = 0x03FFF000

This value has a 32-bit hexadecimal unsigned integer. Each bit represents whether a channel will be used or not. For example, to use a channel between 12 and 15, the 12, 13, 14 and 15 bits of the channel mask should be 1 and all other bits should be 0. The default setting is 0x3FFF000 which means that all the channels from 12 to 25 can be used. The ZE10 supports only the channels between 12 and 25, so 0~11 bits and 26~31 bits should be 0. Table 3-2 shows the bit mask for each channel.

CHANNEL	12	13	14	15	16	17	18
BIT MASK	00001000	00002000	00004000	00008000	00010000	00020000	00040000
CHANNEL	19	20	21	22	23	24	25
BIT MASK	00080000	00100000	00200000	00400000	00800000	01000000	02000000

C. PAN ID CONFIGURATION

Every ZigBee network should be assigned a PAN ID to identify the network. If the PAN ID is 0000, the ZE10 will generate a random 16-bit PAN ID to start the ZigBee network. Otherwise, the user can enter the PAN ID using the AT command as below:

- *AT+PANID = xxxx or AT+PI = xxxx*
where *xxxx* is PAN ID to use. For example, 12AB.

Default setting is 0000.

When choosing a PAN ID, the user needs to make sure that there is no ZigBee network using the same PAN ID. If the PAN ID is already used by another ZigBee network, the PAN ID is unavailable and the user should choose a new PAN ID. In order to avoid this, PAN ID auto-generation is strongly recommended. The user can also assign the 64-bit extended PAN ID to the ZE10 using the AT command below. If the extended PAN ID is

0000000000000000, the ZE10 uses its IEEE address as the extended PAN ID. The extended PAN ID cannot be shared with other ZigBee networks, either.

- *AT+EPID = xxx...xxx or AT+EI = xxx...xxx*
where xxx...xxx is EPID to use. For example, 0123456789ABCDEF
Default setting is 000...000.

D. PERMIT JOINING CONFIGURATION

When a router or an end-device tries to join the ZigBee network, there should be at least one coordinator or router which is permitting joining. If there is no router in the ZigBee network yet, the coordinator should permit joining. The ZE10 provides three ways to permit joining are as follows:

- Permanently permitting by configuration
- Transiently permitting by AT command
- Transiently permitting by digital input signal

If joining is permanently permitted, any device can join anytime. Since this will allow even an unidentified device to join the network anytime, this mode should be selected with extra caution. The AT command to permit joining permanently is as follows:

- *AT+PERMIT = 255 or AT+PJ = 255*

The user can permit joining for a moment at need. The AT command to permit joining is

- *AT+PERMIT = n or AT+PJ = n*
where n is the duration to permit in seconds ranging 1~254 seconds.

The user can also configure the ZE10 to permit joining when the digital input GPIO0 is active. The AT command to enable this functionality is as follows:

- *AT+GPIO0 = 5 or AT+IO0 = 5*

When the ZE10 detects falling edge of the GPIO0, the ZE10 permits joining for the duration that is configured in S-register 22 (S22). The AT command to change the duration is as follows:

- *ATS22 = n*
where n is the duration to permit in seconds ranging 1~254 seconds.

Default is 255.

S22 value determines the time that the device will allow other devices to join to it. If set to 255, the device will always permit other nodes joining. If the user reboots the coordinator (or a router), the device allows other nodes to join the network for the period defined by S-register 22.

Note: *When a device that has joined a network reboots, the device is able to join the network regardless of permitting. But, after the device has joined other network, rejoining the previous network should be permitted by the coordinator or a router.*

EXAMPLE:

AT-COMMAND	RESPONSE	DISCRETION
AT	OK	
AT+NODETYPE?	0 OK	Check the Node type
AT+NODETYPE=1	OK	Set node type as the Coordinator
AT+CHMASK=03FFF000	OK	Set the channel mask
AT+PANID=1234	OK	Set the PAN ID
ATZ	OK	Apply the change
AT+NODETYPE?	1 OK	Check the Node type
AT+CHMASK?	03FFF000 OK	Check the Channel Mask
AT+PANID?	1234 OK	Check the PAN ID No:
AT+EPID?	0000000000000000	Check the Extended PANID

Experiment No.1B

Name

Configuration of the ZigBee Module as a Router

Objective

To study how to configure the ZigBee Module as a Router

Equipments

ZigBee Trainer Kit

Procedure

CONFIGURING ZIGBEE AS ROUTER

A router is a full-function routing device that relays messages and can act as a parent to other nodes. These devices must be always powered on. To configure the ZE10 as a ZigBee router, follow the steps below.

- Configure the ZE10 node type as a router.
- Optionally, select the channel mask. Otherwise, the ZE10 will use the default channel mask.
- Optionally, set up the PAN ID and/or the extended PAN ID. Otherwise, the ZE10 will scan and join the ZigBee network found automatically.
- Permit joining when other router or end-device tries to join the ZigBee network.

A. NODE TYPE CONFIGURATION

The ZE10 module can be configured as a ZigBee router by the AT command as below:

- $AT+NODETYPE = n$ or $AT+NT = n$
where n is 0 = none, 1 = coordinator, 2 = router, 3 = end-device,
4 = sleepy end-device

After the operation mode is selected, ATZ command should be performed to reset the unit to have the change applied.

AT-COMMAND	DESCRIPTION
AT+NODETYPE=2<CR>	Set node type as the router
OK<CR>	Response
ATZ<CR>	Apply the change
OK<CR>	Response

B. CHANNEL MASK CONFIGURATION

When the ZE10 as a ZigBee router tries to join, it scans all the channels configured by channel mask to find the ZigBee network to join. The channel mask of router or end-device should include the channel on which the coordinator is operating. On the assumption that the coordinator of the network to join is operating on channel 12, the router trying to join should have a channel mask of which 12th bit is 1.

C. PAN ID CONFIGURATION

The PAN ID of a ZigBee router or a ZigBee end-device should be the same with the PAN ID of the ZigBee coordinator. If the coordinator is using a randomly generated PAN ID, the user can get the operating PAN ID by running the AT command below on the coordinator:

- AT+OPPANID or AT+OI

If the PAN ID of a ZigBee router is 0000, the PAN ID is ignored. After checking the PAN ID, the ZE10 trying to join compares the extended PAN ID to make sure whether the network is the right network to join. If the extended PAN ID is the same with its configured extended PAN ID, it will try to join the network. Otherwise, it will try to find another network. The user can get the operating extended PAN ID by running the AT command below on the coordinator:

- AT+OPEPID or AT+OE

If the extended PAN ID of a ZigBee router or a ZigBee end-device is 0000000000000000, the extended PAN ID is ignored.

D. PERMIT JOINING CONFIGURATION

The ZigBee router is configured and works the same way with the coordinator to permit joining of other nodes. The ZE10 provides three ways to permit joining are as follows:

- Permanently permitting by configuration
- Transiently permitting by AT command
- Transiently permitting by digital input signal

If joining is permanently permitted, any device can join anytime. Since this will allow even an unidentified device to join the network anytime, this mode should be selected with extra caution. The AT command to permit joining permanently is as follows:

- *AT+PERMIT = 255 or AT+PJ = 255*

The user can permit joining for a moment at need. The AT command to permit joining is

- *AT+PERMIT = n or AT+PJ = n*
where n is the duration to permit in seconds ranging 1~254 seconds.

The user can also configure the ZE10 to permit joining when the digital input GPIO0 is active. The AT command to enable this functionality is as follows:

- *AT+GPIO0 = 5 or AT+IO0 = 5*

When the ZE10 detects falling edge of the GPIO0, the ZE10 permits joining for the duration that is configured in S-register 22 (S22). The AT command to change the duration is as follows:

- *ATS22 = n*
where n is the duration to permit in seconds ranging 1~254 seconds.
Default is 255.

S22 value determines the time that the device will allow other devices to join to it. If set to 255, the device will always permit other nodes joining. If the user reboots the coordinator (or a router), the device allows other nodes to join the network for the period defined by S-register 22.

Note: When a device that has joined a network reboots, the device is able to join the network regardless of permitting. But, after the device has joined other network, rejoining the previous network should be permitted by the coordinator or a router.

EXAMPLE:

AT-COMMAND	RESPONSE	DISCRETION
AT	OK	
AT+NODETYPE?	0 OK	Check the Node type
AT+NODETYPE=2	OK	Set node type as the Router
AT+CHMASK=03FFF000	OK	Set the channel mask
AT+PANID=1234	OK	Set the PAN ID
ATZ	OK	Apply the change
AT+PERMIT=255	OK	Set the Permit
AT+NODETYPE?	2 OK	Check the Node type
AT+CHMASK?	03FFF000 OK	Check the Channel Mask
AT+PANID?	1234 OK	Check the PAN ID No:
AT+EPID?	0000000000000000	Check the Extended PANID

Experiment No.1C

Name

Configuration of the ZigBee Module as a End Device

Objective

To study how to configure the ZigBee Module as a End Device

Equipments

ZigBee Trainer Kit

Procedure

CONFIGURING ZIGBEE AS END DEVICE

An end-device is a leaf node which communicates only with its parent node and will not relay messages directly. To configure the ZE10 as a ZigBee end-device, follow the steps below.

- Configure the ZE10 node type as an end-device
- Optionally, select the channel mask. Otherwise, the ZE10 will use the default channel mask.
- Optionally, set up the PAN ID and/or the extended PAN ID. Otherwise, the ZE10 will scan and join the ZigBee network found automatically.
- Configure the poll timeout and the poll period

1. NODE TYPE CONFIGURATION

The ZE10 module can be configured as a ZigBee router by the AT command as below:

- *AT+NODETYPE = n or AT+NT = n*
where n is 0 = none, 1 = coordinator, 2 = router, 3 = end-device,
4 = sleepy end-device

After the operation mode is selected, ATZ command should be performed to reset the unit to have the change applied

AT-COMMAND	DESCRIPTION
AT+NODETYPE=3<CR>	Set node type as the end-device
OK<CR>	Response
ATZ<CR>	Apply the change
OK<CR>	Response

2. C

HANNEL MASK CONFIGURATION

When the ZE10 as a ZigBee end-device tries to join, it scans all the channels configured by channel mask to find the ZigBee network to join. The channel mask of router or end-device should include the channel on which the coordinator is operating. On the assumption that the coordinator of the network to join is operating on channel 12, the end-device trying to join should have a channel mask of which 12th bit is 1.

3. PAN ID CONFIGURATION

The PAN ID of a ZigBee end-device should be the same with the PAN ID of the ZigBee coordinator. If the coordinator is using a randomly generated PAN ID, the user can get the operating PAN ID by running the AT command below on the coordinator:

- `AT+OPPANID` or `AT+OI`

If the PAN ID of a ZigBee end-device is 0000, the PAN ID is ignored. After checking the PAN ID, the ZE10 trying to join compares the extended PAN ID to make sure whether the network is the right network to join. If the extended PAN ID is the same with its configured extended PAN ID, it will try to join the network. Otherwise, it will try to find another network. The user can get the operating extended PAN ID by running the AT command below on the coordinator:

- `AT+OPEPID` or `AT+OE`

If the extended PAN ID of a ZigBee router or a ZigBee end-device is 0000000000000000, the extended PAN ID is ignored.

4. PERMIT JOINING CONFIGURATION

The ZigBee end-device is configured and works the same way with the coordinator to permit joining of other nodes. The ZE10 provides three ways to permit joining are as follows:

- Permanently permitting by configuration
- Transiently permitting by AT command
- Transiently permitting by digital input signal

If joining is permanently permitted, any device can join anytime. Since this will allow even an unidentified device to join the network anytime, this mode should be selected with extra caution. The AT command to permit joining permanently is as follows:

- `AT+PERMIT = 255` or `AT+PJ = 255`

The user can permit joining for a moment at need. The AT command to permit joining is

- `AT+PERMIT = n` or `AT+PJ = n`
where n is the duration to permit in seconds ranging 1~254 seconds.

The user can also configure the ZE10 to permit joining when the digital input GPIO0 is active. The AT command to enable this functionality is as follows:

- `AT+GPIO0 = 5` or `AT+IO0 = 5`

When the ZE10 detects falling edge of the GPIO0, the ZE10 permits joining for the duration that is configured in S-register 22 (S22). The AT command to change the duration is as follows:

- $ATS22 = n$
where n is the duration to permit in seconds ranging 1~254 seconds.
Default is 255.

S22 value determines the time that the device will allow other devices to join to it. If set to 255, the device will always permit other nodes joining. If the user reboots the coordinator, the device allows other nodes to join the network for the period defined by S-register 22.

Note: When a device that has joined a network reboots, the device is able to join the network regardless of permitting. But, after the device has joined other network, rejoining the previous network should be permitted by the coordinator or a router.

When an end-device joins the network, the power LED blinks periodically if GPIO1 is set to specific functionality. The end-device belongs to a parent node which is either the coordinator or a router, and the parent node registers the end-device to its internal child table for message relaying. This table is maintained as long as the end-device polls the parent node periodically within the poll timeout value of the parent node. If the end-device does not poll for the poll timeout value of the parent node, the parent node removes the end-device from its child table and does not relay the message any more. Hence, the poll period of the end-device should be equal or shorter than the poll timeout of the parent node to avoid accidental removal of the end-device from the parent node's child table.

The ZE10 poll interval can be configured by setting the S-registers using the AT commands as below.

- $ATS51 = n$
where n is poll timeout in seconds and the default is 60.

This value is for a parent node. If an end-device does not send a poll request to its parent coordinator or router within the poll timeout, the end-device is removed from the child table.

- $ATS56 = n$
where n is poll period in seconds and the default is 5.

End-device polls the parent node at poll period defined by S56 periodically. It is for the connection to the parent node and data transmission with other nodes in the network.

- $ATS55 = n$
where n is rejoin timeout and the default is 10.

When the connection to the parent node is broken for S55 in seconds, the child node tries to rejoin the current network.

Even though the end-device needs to poll the parent node continuously, it does not go into the sleep mode and it can send and receive messages any time unlike the sleepy end-device.

EXAMPLE:

AT-COMMAND	RESPONSE	DISCRETION
AT	OK	
AT+NODETYPE?	0 OK	Check the Node type
AT+NODETYPE=3	OK	Set node type as the End-Device
AT+CHMASK=03FFF000	OK	Set the channel mask
AT+PANID=1234	OK	Set the PAN ID
ATZ	OK	Apply the change
AT+PERMIT=255	OK	Set the Permit
AT+NODETYPE?	3 OK	Check the Node type
AT+CHMASK?	03FFF000 OK	Check the Channel Mask
AT+PANID?	1234 OK	Check the PAN ID No:
AT+EPID?	0000000000000000	Check the Extended PANID

Experiment No. 2

Experiment No.2

Name

Communication with two ZigBee Module

Objective

To Study how to Communicate with two ZigBee Module

1. Data Communication between Coordinator and Router Module
2. Data Communication between Coordinator and End Device

Equipments

- ZigBee Trainer Kit - 02
- USB Cable - 02
- Power Supply - 02

Procedure

COMMUNICATION WITH TWO ZIGBEE MODULE

The ZE10 supports three different types of data transmissions: unicast, multicast and broadcast. Unicast is to transmit data to a specific ZigBee device. Multicast is to transmit data to a group of ZigBee devices that are assigned a specific group ID. Broadcast is to transmit data to all ZigBee devices (or specific type of ZigBee nodes if specified) in the network.

ZE10 has such operation modes as ***command mode and data mode***. Data can be transmitted at each operation mode as below:

- **COMMAND MODE:** This mode is for entering commands to set and manage a ZE10 ZigBee Modem, as many data as allowed are transmitted at a time by using AT command.
- **DATA MODE:** This mode is for transmitting data, it is possible to send data consecutively. “+++” changes the operation mode from the data mode to command mode.

COMMAND MODE

UNICAST

Unicast transmissions are sent from a source device to a destination device. The destination device can be any ZigBee device in the ZigBee network. To specify the destination device, the ZigBee address of the destination device should be known.

There are two different ZigBee addresses assigned to each ZigBee device. The 64-bit address is a unique device address assigned during manufacturing. The 64-bit address is also called IEEE address or extended address. The 16-bit address is assigned to the device when it joins a ZigBee network. The ZE10 can send unicast messages based on either 16-bit address or 64-bit address. The AT command for the unicast transmissions is as below:

- `AT+UNICAST = <node_id>,<MSG><CR> o`
- `AT+UC = <node_id>,<MSG><CR>`

where <node_id> = 16-bit or 64-bit address, <MSG> = data to transmit

The maximum payload of the message is 90 bytes (or 72 bytes under security enabled). If the message length exceeds the maximum payload, the message cannot transmit and the error messages will be presented. The AT command to get the maximum payload is as below:

- `AT+MAXPAYLOAD?<CR>`
- `AT+MP?<CR>`

MULTICAST

Multicast transmissions are sent from a source device to destination devices with the specific group ID. The group ID is assigned to the devices using the AT+GROUPID command. The AT command for the multicast transmissions is as below.

- `AT+MULTICAST=<group_id>,<MSG><CR>`
- `AT+MC=<group_id>,<MSG><CR>`

where <group_id> = group ID, <MSG> = data to transmit

The data transmission range can be set using the AT commands below. Please refer to the **S-Registers** for more details.

ATS36=n, where n is the radius.

ATS37=n, where n is the number of hops through nonmembers.

Note: *ZigBee compliance rules prevents flooding of the network by limiting the network to a maximum of 9 broadcasts over 8 seconds. Attempts to submit additional broadcasts beyond this should result in the transmission error. This is approximately one broadcast per second. It is, however, recommended to broadcast one packet over 2 seconds for safe transmission. Broadcast transmission failure also causes retransmission which is affected by transmission timeout (S33) and transmission retries (S32). Because frequent retransmissions may cause to break the broadcast limitation, it is recommended to set more than 1000 for transmission timeout (S33) and set as small as possible for transmission retries(S32) . Multicast data transmission is implemented on top of the network broadcast mechanism, so it is subject to the same rules.*

BROADCAST

Broadcast transmissions are sent from a source device to all devices in the ZigBee network. The AT command for the broadcast transmissions is as below.

- `AT+BROADCAST=<MSG><CR>`

- **AT+BC=<MSG><CR>**
where <MSG> = data to transmit

The user can also specify the transmission range by changing the S35 register value as below. Please refer to the **S-Registers** for more details.

ATS35=n, where n is the broadcast range.

ATS36=n, where n is the radius.

Note: ZigBee compliance rules prevent flooding of the network by limiting the network to a maximum of 9 broadcasts over 8 seconds. Attempts to submit additional broadcasts beyond this should result in the transmission error. This is approximately one broadcast per second. It is, however, recommended to broadcast one packet over 2 seconds for safe transmission. Broadcast transmission failure also causes retransmission which is affected by transmission timeout (S33) and transmission retries (S32). Because frequent retransmissions may cause to break the broadcast limitation, it is recommended to set more than 1000 for transmission timeout (S33) and set as small as possible for transmission retries (S32). Multicast data transmission is implemented on top of the network broadcast mechanism, so it is subject to the same rules.

SENDING BINARY DATA

Because some characters like carriage return, line feed and back space collides with AT commands in command mode and back slash('\') is used for control character for binary encoding, they cannot be transmitted as they are. The characters out of range between 0x20(Space) and 0x7E('~') in hexadecimal are not expressed at the AT commands for transmitting data. In order to transmit these characters in command mode, these characters must be encoded in a command line.

For example, when sending *abc<CR>* with AT+BROADCAST command, the command should be as follows:

- **AT+BROADCAST=abc\0D<CR>**

DISPLAY INCOMING MESSAGE

S11 register determines whether incoming message will be shown or not in the command mode. In the data mode, on the other hand, incoming data is always displayed. To change the S11 register, please use the AT command as below:

- **ATS11=<value><CR>**
where <value> = 0 to hide incoming message, 1 to show incoming message

If user wants to display incoming message in command mode, S11 should be enabled. It is displayed by the form of **+<IEEE address>|<MSG><CR>**.

When displaying incoming messages in command mode, some characters are encoded in the same way with sending binary data.

For example, when receiving *abc<CR>* from 0001950000000001, the displaying message is as follows:

+0001950000000001| abc\0D<CR>

Note: In PTV1.2 or earlier version, only the characters mentioned at Table 4-1 are encode. In PTV1.3 or later version, back slash ('\') and the characters out of range between 0x20(Space) and 0x7E (~') in hexadecimal are encoded.

DATA MODE

If the data is too long to send using the command mode, Data Mode can be used instead. Once the ZE10 enters a Data Mode session, the ZE10 can send data continuously to a specific node or current network until the data mode is escaped. The data is packetized and transmitted at every inter character timeout.

This data link to the designated device is maintained until the escape command “+++” is entered. To return to the last data mode after the ZE10 is escaped by “+++”, use the AT command as below.

- *ATO<CR>*

UNICAST DATA MODE

Unicast data mode can be used for continuous data transmission to a particular node. Unicast data mode can be entered using either ATD or AT+TRANSMITMODE=1 (or 4).

- ATD command is the command for entering the unicast data mode.
- AT+TRANSMITMODE=1 is used to set the data transmission mode to unicast to destination long address
- AT+TRANSMITMODE=4 to unicast to sink node.

If ATD command is used, the device will enter the unicast data mode immediately and keep it until escaped using “+++” or reset by ATZ. If the device is reset, it will not enter the unicast data mode but command mode in this case.

On the other hand, if AT+TRANSMITMODE=1 (or 4) is used to enter the unicast data mode, the device enters the unicast data mode again whenever the device is reset. To disable the unicast data mode, escape the data mode first by using “+++” string and use AT+TRANSMITMODE=0 and reset the device to apply the setting permanently.

AT commands used for the unicast data mode are as below. Set the destination address which is used for ATD command or AT+ TRANSMITMODE=1.

- *AT+DESTLA=<value> or AT+DL=<value>*
where <value> is 64-bit IEEE address of destination node for unicast data mode.

Set unicast data mode to the preconfigured destination IEEE address without reset.

- *ATD<CR>*

Set unicast data mode to <value> device and <value> can be either 16-bit node ID or 64-bit IEEE address of remote node.

- *ATD<value><CR>*

Set the transmission mode to unicast to a destination IEEE address. The device should be reset to apply the transmission mode.

- **AT+TRANSMITMODE=1<CR> or AT+TM=1<CR>**

The 'unicast to sink node data mode' is useful for a device to gather data from many devices because every device comes to know the destination device by setting the destination device as sink node instead of describing the destination device to every device. Every device also comes to know the new destination device by changing the sink node in the network.

AT commands used for the 'unicast to sink node data mode' are as below.

- **ATS61=<value><CR>**
where <value> = 0 not sink node, <value> > 0 sink node to advertise its address to the devices in the destination group (AT+DESTGROUPID) every <value> second.
- **AT+DESTGROUPID=<value> or AT+DG=<value>**
where <value> is 16-bit destination group ID to which sink node advertises its address.

Set destination Group ID of the device.

- **AT+SINKINFO?<CR> or AT+SI?<CR>**

Get the long and short addresses of sink node. If there is no sink node in the network, the addresses will be "FFFFFFFFFFFFFFFF" and "FFFF".

- **AT+TRANSMITMODE=4<CR> or AT+TM=4<CR>**

Set the transmission mode to unicast to a sink node. The device should be reset to apply the mode.

AT-COMMAND	RESPONSE	DISCRETION
AT+DESTLA=00019500002FDC40	OK	Set the destination IEEE address
ATD	OK	Enter the unicast mode using destination IEEE address
SEND UNICAST MESSAGE		Unicast message
+++	OK	Leave data mode

AT-COMMAND	RESPONSE	DISCRETION
ATD0000	OK	Enter the unicast mode using 16-bit node ID
SEND UNICAST MESSAGE		Unicast message
+++	OK	Leave data mode

AT-COMMAND	RESPONSE	DISCRETION
AT+DESTLA=00019500002FDC40	OK	Set the destination IEEE address
SEND UNICAST MESSAGE		Unicast message
+++	OK	Leave data mode

AT-COMMAND	RESPONSE	DISCRETION
AT+DESTLA=00019500002FDC40	OK	Set the destination IEEE address
AT+TRANSMITMODE=1	OK	Set transmission mode to destination IEEE address
ATZ	OK	Apply to change
SEND UNICAST MESSAGE		Unicast Message

+++	OK	Leave data mode
-----	----	-----------------

AT-COMMAND	RESPONSE	DISCRETION
AT+TRANSMITMODE=4	OK	Set transmission mode to destination IEEE address
ATZ	OK	Apply to change
SEND UNICAST MESSAGE		Unicast Message
+++	OK	Leave data mode

MULTICAST DATA MODE

Multicast data mode can be used for continuous data transmission to a particular group of multiple devices. Multicast data mode can be entered using either ATM or AT+TRANSMITMODE=2. If ATM command is used, the device will enter the multicast data mode immediately and keep it until escaped using “+++” or reset by ATZ. If the device is reset, it will not enter the multicast data mode but command mode in this case.

On the other hand, if AT+TRANSMITMODE=2 is used to enter the multicast data mode, the device enters the multicast data mode again whenever the device is reset. To disable the multicast data mode, escape the data mode first by using “+++” string and use AT+TRANSMITMODE=0 and reset the device to apply the setting permanently.

AT commands used for the multicast data mode are as below.

Set Group ID of the device. The device will receive the multicast data sent to this Group ID.

- *AT+GROUPID=<value> or AT+GI=<value>, where <value> is 16-bit group ID for multicast data mode.*

Set destination Group ID of the device.

- *AT+DESTGROUPID=<value> or AT+DG=<value>, where <value> is 16-bit destination group ID for multicast data mode.*

Set multicast data mode to preconfigured destination group ID without reset.

- *ATM<CR>*

Set multicast data mode to <value> group ID and <value> can be 16-bit ID.

- *ATM<value><CR>*

Set the transmission mode to multicast mode. The device should be reset to apply the mode.

- *AT+TRANSMITMODE=2<CR> or AT+TM=2<CR>*

Set data relay and forwarding options. Please refer to the **S-Registers** for more details.

- *ATS36=n, where n is the radius.*
- *ATS37=n, where n is the number of hops through nonmembers.*

AT-COMMAND	RESPONSE	DISCRETION
ATM0001	OK	Enter the multicast mode using 16-bit group ID

SEND UNICAST MESSAGE		Unicast Message
+++	OK	Leave data mode

AT-COMMAND	RESPONSE	DISCRETION
AT+DESTGROUPID=00019500002FDC40	OK	Set the destination IEEE address
ATM	OK	Enter the multicast mode using destination group ID
SEND UNICAST MESSAGE		Unicast Message
+++	OK	Leave data mode

AT-COMMAND	RESPONSE	DISCRETION
AT+DESTGROUPID=0001	OK	Set the destination group ID
AT+TRANSMITMODE=2	OK	Set transmission mode to destination group ID
ATZ	OK	Apply to change
SEND UNICAST MESSAGE		Unicast Message
+++	OK	Leave data mode

BROADCAST DATA MODE

Broadcast data mode can be used for continuous data transmission to all devices in the current network. The destination address or group ID of the remote nodes is not necessary for broadcast data mode. The transmit range is determined by the S-register 35

Broadcast data mode can be entered using either ATB or AT+TRANSMITMODE=3. If ATB command is used, the device will enter the broadcast data mode immediately and keep it until escaped using “+++” or reset by ATZ. If the device is reset, it will not enter the broadcast data mode but command mode in this case.

On the other hand, if AT+TRANSMITMODE=3 is used to enter the broadcast data mode, the device enters the broadcast data mode again whenever the device is reset. To disable the broadcast data mode, escape the data mode first by using “+++” string and use AT+TRANSMITMODE=0 and reset the device to apply the setting permanently.

- AT commands used for the broadcast data mode are as below.

Set broadcast data mode.

- *ATB<CR>*

Set the transmission mode to broadcast. It should be reset to apply the mode.

- *AT+TRANSMITMODE=3<CR> or AT+TM=3<CR>*

- *ATS35=n, where n is the broadcast range.*

- *ATS36=n, where n is the radius.*

Set data transmission range. Please refer to the **S-Registers** for more details.

AT-COMMAND	RESPONSE	DISCRETION
ATS35=2	OK	Set broadcast range to all nodes in the network
ATB	OK	Enter the broadcast mode
SEND UNICAST MESSAGE		Unicast Message
+++	OK	Leave data mode

AT-COMMAND	RESPONSE	DISCRETION
AT+TRANSMITMODE=3	OK	Set transmission mode to current network
ATZ	OK	Apply to change
SEND UNICAST MESSAGE		Unicast Message
+++	OK	Leave data mode

Experiment No.2A

Name

Communication with Coordinator and Router Module

Objective

To Study how to do Data Communication between Coordinator and Router Module

Equipments

ZigBee Trainer Kit - 02

Procedure

DATA COMMUNICATION BETWEEN COORDINATOR MODULE AND ROUTER MODULE

1. Connect trainers to the PC and turn off all on board switches then Power On the trainer.
2. Run ProBee Manger separately for two modules.
3. Configure the Serial port (Baud - 9600, Data - 8, Parity - None, Stop Bit-1, Flow control - None).
4. Click OPEN icon at ProBee manger, to connect ZigBee module to PC
5. Press RESET TO DEFAULT ICON and reload the ZigBee using RELOAD icon.
6. Go to Terminal window and follow the AT-Commands
7. While sending data enable Local Echo.

COORDINATOR	ROUTER
AT OK AT+NODENAME=COORDINATOR OK AT+NODETYPE=1 OK AT+PANID=ABCD OK ATS11=1 OK ATZ OK OK -----# SET THE ROUTER AT+NODETYPE? 1 OK	AT OK AT+NODENAME=ROUTER OK AT+NODETYPE=2 OK AT+PANID=ABCD OK ATS11=1 OK ATZ OK OK ----- # CHECK AT-CMD IN BOTH SIDE AT+NODETYPE? 2 OK

AT+NODENAME? COORDINATOR OK AT+PANID? ABCD OK AT+CHMASK? 03FFF000 OK AT+PERMIT=255 OK AT+REMOTE=0001950000004D9E,AT+SHORTA DDR F4C2	AT+NODENAME? ROUTER OK AT+PANID? ABCD OK AT+CHMASK? 03FFF000 OK AT+PERMIT=255 OK AT+LONGADDR 0001950000004D9E OK
--	---

COORDINATOR	ROUTER
ATDF4C2 OK OK FROM COORDINATOR // DATA TO ROUTER FROM ROUTER // DATA FROM ROUTER +++ OK	AT+SHORTADDR F4C2 OK +0001950000004D9E F +0001950000004D9E R +0001950000004D9E O +0001950000004D9E M +0001950000004D9E +0001950000004D9E C +0001950000004D9E O +0001950000004D9E O +0001950000004D9E R +0001950000004D9E D +0001950000004D9E I +0001950000004D9E N +0001950000004D9E A +0001950000004D9E T +0001950000004D9E O +0001950000004D9E R AT OK ATD0000 OK FROM ROUTER // DATA TO CODTR +++ OK

Experiment No.2B

Name

Communication with Coordinator Module and End Device

Objective

To Study how to do Data Communication between Coordinator Module and End Device

Equipments

ZigBee Trainer Kit - 02

Procedure

DATA COMMUNICATION BETWEEN COORDINATOR MODULE AND END DEVICE MODULE

1. Connect trainers to the PC and turn off all switches then Power On the trainer.
2. Run ProBee Manger separately for two modules.
3. Configure the Serial port (Baud - 9600, Data - 8, Parity - None, Stop Bit-1, Flow control - None).
4. OPEN icon in ProBee manger is used to connect ZigBee module to PC.
5. Press RESET TO DEFAULT ICON and reload the ZigBee using RELOAD icon.
6. Go to Terminal window and follow the AT-Commands
7. While sending data enable Local Echo.

COORDINATOR	END DEVICE
AT OK AT+NODETYPE=1 OK AT+NODENAME=COORDINATOR OK AT+PANID=ABCD OK ATS11=1 OK ATZ OK OK ----- # SET THE END DEVICE AT+NODETYPE? 1	AT OK AT+NODETYPE=3 OK AT+NODENAME=END DEVICE OK AT+PANID=ABCD OK ATS11=1 OK ATZ OK OK ----- # CHECK AT-CMD IN BOTH SIDE AT+NODETYPE? 3

OK AT+NODENAME? COORDINATOR OK AT+PANID? ABCD OK AT+CHMASK? 03FFF000 OK AT+PERMIT=255 OK AT+REMOTE=0001950000004D9E,AT+SHORT F4C2 ATDF4C2 OK OK FROM COORDINATOR // DATA TO E-DEVICE FROM END DEVICE // DATA FROM E-DEVICE +++ OK	OK AT+NODENAME? END DEVICE OK AT+PANID? ABCD OK AT+CHMASK? 03FFF000 OK AT+PERMIT=255 OK AT+LONGADDR 0001950000004D9E OK AT+SHORTADDR F4C2 OK +0001950000004D95 F +0001950000004D95 R +0001950000004D95 O +0001950000004D95 M +0001950000004D95 +0001950000004D95 C
COORDINATOR	ROUTER
	+0001950000004D95 O +0001950000004D95 O +0001950000004D95 R +0001950000004D95 D +0001950000004D95 I +0001950000004D95 N +0001950000004D95 A +0001950000004D95 T +0001950000004D95 O +0001950000004D95 R AT OK ATD0000 OK FROM END DEVICE // DATA TO CODTR +++ OK

Experiment
No. 3

Experiment No.3

Name

Interfacing the Peripherals

Objective

To study how to interface the peripherals

1. On Board peripherals
2. Remote Board peripherals

Equipments

- ZigBee Trainer Kit
- USB Cable
- Power Supply

Procedure

ZIGBEE PERIPHERAL INTERFACING

The ZigBee Trainer has different types of on board peripherals like temperature sensor, light sensor and it also have Buzzer, Relay, LEDs and Switches. All the peripherals are user accessible through AT-Command only. Peripherals are rooted internally; so that the ZigBee module's specific GPIO pins are connected to a particular device.

Refer the ZIGBEE Hardware details for more about Peripherals configuration.

Experiment No.3

Name

Interfacing the On Board Peripherals

Objective

To study how to interface On Board peripherals

Equipments

ZigBee Trainer Kit

Theory

INTERFACING ON BOARD PERIPHERALS

The ZE10 module has total 14 GPIO pins. Among these pins, 6 GPIO pins are assigned to factory reset only and the user can use the rest 13 pins for general purpose inputs and outputs.

The AT commands used to change the GPIO settings.

- AT+GPIO=<values><CR> or AT+GPIO<number>=<value><CR>
where <values> is a number between 0~5 as explained below:
 - 0 > Disabled (unmonitored digital input)
 - 1 > Digital input, monitored
 - 2 > Digital output, default low
 - 3 > Digital output, default high
 - 4 > Analog input, single ended (only valid for GPIO3~6)
 - 5 > Reserved for pin-specific alternate functionalities

The GPIO status can be monitored and controlled using the AT commands. The GPIO status of remote device can be monitored and controlled using remote AT commands, too. The ZE10 can also be configured to sample the GPIO status and send to a remote device periodically.

QUERIED SAMPLING

The GPIO status can be monitored and controlled using the AT commands either locally or remotely. To read the digital GPIO line values, the AT+DIO command can be used as below. Return value is a combination of 0 (Low) and 1 (High) if the pins are configured as digital GPIO lines. If the pins are configured as analog inputs, the corresponding return values are shown

- AT+DIO?<CR> or AT+DIO<number>?<CR>
where <number> is the specific IO number between ranges 0 ~ 12 to read

AT+DIO command can also be used to set the digital output lines.

- *AT+DIO=<value><CR> or AT+DIO<number>=<value><CR>*
where <number> is the specific IO number between ranges 0 ~ 12 to read, and the<values> is a combination of 0 or 1, 0 = Low, 1 = High

The analog input values can be read using the AT+AI command. Return value is a combination of 16-bit hexadecimal integer between 0~2EE0(0~1200mV) or D120-FFFF(-1200~-1mV) which unit value corresponds to 0.1mv. To convert the hexadecimal value to the actual voltage, please use the formula below.

- *AT+AI?<CR> or AT+AI<number>?<CR>*
where <number> is the specific IO number between ranges 0 ~ 3 to read
 - *If the Value is between 0 ~ 2EE0, Voltage (mV) = Value * 0.1*
 - *If the Value is between D120 ~ FFFF, Voltage (mV) = (Value – 65536) * 0.1*

PERIODIC SAMPLING

The ZE10 can be configured to send its GPIO status to a remote device at a periodic rate.

- *AT+DESTLA=<IEEE address><CR> or AT+DL=<IEEE address><CR>*
where <IEEE address> is the long address of a remote node to send sampled

GPIO values of local node.

- *ATS42=<value><CR>*
where <value> is the GPIO sampling period in seconds. If set to 0, GPIO monitoring is disabled.

- *ATS43=<value><CR>*
where <value> is the GPIO monitoring node. If set to 0, the monitoring node is selected by AT+DESTLA. If set to is 1, the sink node is set as the monitoring node.

A device can be a sink node by setting S-register 61. The sink node multicasts its address to the other devices in the destination group in the network according to the value of S-register 61, which is called advertising.

- *ATZ<CR>*

Reset the device to apply the changes

The ZE10 can be configured to display periodic sampling data coming from the other nodes. To enable this feature, please use the AT command below:

- *ATS11=<value><CR>*
where <value> determines whether incoming message will be presented or not. If a user wants to display incoming message in command mode, S11 should be enabled.

Display message format: ++<IEEE address>|<GPIO value><CR>.

CHANGE DETECTION SAMPLING

The ZE10 module can take an I/O sample and transmit it to a remote device immediately whenever a monitored digital I/O pin changes state. It is possible to configure it with periodic sampling simultaneously.

- *AT+DESTLA=<IEEE address><CR>*

where <IEEE address> is the long address of a remote node to send sampled GPIO values of local node.

- **ATS43=<value><CR>**
where <value> is the GPIO monitoring node. If set to 0, the monitoring node is selected by AT+DESTLA. If set to 1, the sink node is set as the monitoring node.
- **ATS44=<value><CR>**
Set GPIO change detection sampling mode. If <value> is set to 0, GPIO sampling is disabled.
- **ATZ<CR>**
Reset the device to apply the changes

GPIO REMOTE CONTROL

The ZE10 module supports that a local node can change the digital output status by switching digital input pins of a remote node. To use this function, it should be possible for the local node to monitor the GPIO status of the remote node (GPIO monitoring period, S42 > 0) or to detect the GPIO change of the remote node (GPIO change detection sampling mode, S44 = 1).

To configure this way, please use the AT commands below:

- **AT+SOURCELA=<IEEE address><CR> or AT+SL=<IEEE address><CR>**
where <IEEE address> is the long address of a specific source node for changing the status of digital output pin. If this address is set to "FFFFFFFFFFFFFFFF", a local node allows that any node can control the GPIO status.
- **ATS45=<value><CR>**
Set GPIO remote control mode. If <value> is set to 0, GPIO control is disabled.

Note: If the GPIO_IN (SW19) is set to Active High, GPIO pull up/down option should be set to Pull-Down (ATS41=0). If the GPIO_IN (SW19) is set to Active Low, GPIO pull up/down option should be set to Pull-Up. (ATS41=1)

Procedure

1) INTERFACING WITH LEDS AND SWITCHS.

In OUTPUT LED units have three LEDs that can accessible by the user. Initially all LEDs are active low state if you want to enable a particular LED then you need to set that GPIO pin (group) as digital output.

LED NAME	ZigBee PIN	GPIO	LED-OFF	LED-ON
LED1	PIN-05	GPIO_0	ACTIVE LOW	ACTIVE HIGH
LED2	PIN-24	GPIO_7	ACTIVE LOW	ACTIVE HIGH
LED3	PIN-11	GPIO_6	ACTIVE LOW	ACTIVE HIGH

Active High = Logical One
Active Low = Logical Zero

AT-Command used to configure the LED GPIO pins output mode.

AT-COMMAND	RESPONSE	DISCRETION
AT+GPIO7=3	OK	Set The GPIO Function
ATZ	OK	Apply to change
AT+DIO7=0	OK	OFF LED2
AT+DIO7=0	OK	ON LED2

In SWITCH units have three SLIDING SWITCH that can accessible by the user. Initially all SWITCH are active low state and it act as input device. If you want to access a particular (group) SWITCH as input device first enable the GPIO as digital input mode then we can access those pin as input device.

SWITCH	ZigBee PIN	GPIO	SWITCH-OFF	SWITCH-ON
SW1	PIN-23	GPIO_9	ACTIVE LOW	ACTIVE HIGH
SW2	PIN-22	GPIO_8	ACTIVE LOW	ACTIVE HIGH
SW3	PIN-19	GPIO_10	ACTIVE LOW	ACTIVE HIGH

Active High = Logical One
Active Low = Logical Zero

AT-Command used to configure the SWITCH GPIO pins output mode.

AT-COMMAND	RESPONSE	DISCRETION
AT+GPIO9=1	OK	Set The GPIO Function
ATZ	OK	Apply to change
AT+DIO9?	1 OK	SW2 ON
AT+DIO9?	0 OK	SW2 OFF

2) INTERFACING WITH RELAY & BUZZER.

RELAY is an output units and it is controlled by ZE10 modem using AT-Commands. It has on board indicator to study the operation of relay and a Terminal connector to connect external device. Trainer provides jumper connection to set these options. If you want to work with relay then you need to set that GPIO pin (group) as digital output.

RELAY	ZigBee PIN	GPIO	ON	OFF
	PIN-18	GPIO_11	ACTIVE HIGH	ACTIVE LOW

JUMPER	JUMPER PINS			RELAY CONNECTION
	1	2	3	
J3	CONNECTED			LED-NO
J4	CONNECTED			+3.3V
J5	CONNECTED			LED-NC
J3		CONNECTED		NO
J4		CONNECTED		COMMON
J5		CONNECTED		NC

AT-Command used to configure the RELAY GPIO pins output mode.

AT-COMMAND	RESPONSE	DISCRETION
AT+GPIO11=3	OK	Set The GPIO Function
ATZ	OK	Apply to change
AT+DIO11=0	OK	RELAY-OFF
AT+DIO11=1	OK	RELAY-ON

In BUZZER units is also an output unit. To access the buzzer need to enable GPIO as digital output mode.

BUZZER	ZigBee PIN	GPIO	ON	OFF
	PIN-17	GPIO_12	ACTIVE HIGH	ACTIVE LOW

AT-Command used to configure the BUZZER GPIO pins output mode.

AT-COMMAND	RESPONSE	DISCRETION
AT+GPIO12=3	OK	Set The GPIO Function
ATZ	OK	Apply to change
AT+DIO12=0	OK	BUZZER OFF
AT+DIO9?	OK	BUZZER ON

3) INTERFACING WITH ADC USING VARIABLE RESISTOR

It is used to study the ADC feature of ZE10 Modem. In ADC section a 10k potentiometer is use to varying voltage. If you want to access the ADC first enable GPIO pin as Analog input mode. ADC gives hexadecimal quantities only, manually we need to convert it into corresponding Voltage parameters.

SENSOR	VOLTAGE	ZigBee PIN	GPIO
ADC	0 - 1.21V	PIN-10	GPIO_5/ADC_2

Voltage across the LDR is high that means LUX is low and wise versa.

AT-Command used to configure the GPIO as Analog Input.

AT-COMMAND	RESPONSE	DISCRETION
AT+GPIO5=4	OK	Set The GPIO Function
ATZ	OK	Apply to change
AT+AI2?	2AF8 OK	Response

Data Conversion

Analog Data Format

Analog data from illumination sensor is a combination of 16-bit hexadecimal integer between 0~2EE0(0~1200mV) which unit value corresponds to 0.1 mV.

Conversion of Analog Data into Voltage

- The Voltage can calculate from Hexadecimal to Decimal conversation.
- Decimal quaintly into millivolt.
- Voltage = (Decimal X .1) / 1000

Example

HEX = 2AF8

Decimal = 11000

Voltage = (11000 X 0.1) / 1000 = 1.1V

4) INTERFACING WITH LDR SENSOR

It is used to measure the Light intensity from 0 Lux to 300 Lux. The sensor is connected to ADC pin of ZigBee module. If you want to access the ADC first enable GPIO pin as Analog input mode. ADC gives hexadecimal quantities only, manually we need to convert it into corresponding illumination parameters.

SENSOR	VALUE	VOLTAGE	ZigBee PIN	GPIO
LDR	0 Lux to 300 Lux	0 - 1.21V	PIN-08	GPIO_4/ADC_1

Voltage across the LDR is high that means LUX is low and vice versa.

AT-Command used to configure the GPIO as Analog Input.

AT-COMMAND	RESPONSE	DISCRETION
AT+GPIO4=4	OK	Set The GPIO Function
ATZ	OK	Apply to change
AT+AI1?	00D2 OK	Response

Data Conversion

Analog Data Format

Analog data from illumination sensor is a combination of 16-bit hexadecimal integer between 0~2EE0(0~1200mV) which unit value corresponds to 0.1 mV.

Conversion of Analog Data into Illumination

- The Vout can calculate from Hexadecimal to Decimal conversation.
- Decimal quantity into millivolt.
- The general equation used to convert output voltage to Illumination is:
- $\text{Illumination (Lux)} = (1200 - \text{Volt (mV)}) \times 0.25 \text{ (Lux/mV)}$

Example

HEX = 00D2

Decimal = 210

Volt (mV) = $210 \times 0.1 = 21\text{mV}$

Lux = $1200 - 21 \times 0.25 = 294.5\text{Lux}$

5) INTERFACING WITH TEMPERATURE SENSOR

It is used to measure the temperature ranging from 0°C to 120°C. The sensor is connected to ADC pin of ZigBee module. If you want to access the ADC first enable GPIO pin as Analog input mode. ADC gives hexadecimal quantities only, manually we need to convert it into corresponding temperature parameters.

SENSOR	VALUE	VOLTAGE	ZigBee PIN	GPIO
TEMP	0°C to 120°C	0 - 1.21V	PIN-07	GPIO_3/ADC_0

AT-Command used to configure the GPIO as Analog Input.

AT-COMMAND	RESPONSE	DISCRETION
AT+GPIO3=4	OK	Set The GPIO Function
ATZ	OK	Apply to change
AT+AI0?	0D23 OK	Response

Data Conversion

Analog Data Format

Analog data from temperature sensor is a combination of 16-bit hexadecimal integer between 0 ~ 2EE0 (0 ~ 1200mV) which unit value corresponds to 0.1 mV.

Conversion of Analog Data into Temperature

- The Vout can calculate from Hexadecimal to Decimal conversation.
- The sensor has a sensitivity of 10mV/°C.
- Use a conversion factor that is the reciprocal that is 100 °C/V.
- The general equation used to convert output voltage to temperature is:
- $\text{Temperature (}^{\circ}\text{C)} = \text{Vout} / (100 \text{ }^{\circ}\text{C} / \text{V})$
- So if Vout is 1V , then, Temperature = 100°C
- The output voltage varies linearly with temperature.

Example

HEX = 0D23

Decimal = 3363

$\text{Vout} = (3363 \times 0.1) / 1000$

$\text{Temp (}^{\circ}\text{C)} = 3363 \times 100 = 33.63^{\circ}\text{C}$

Experiment No.3B

Name

Interfacing the Remote Peripherals

Objective

To study how to interface Remote peripherals

Equipments

ZigBee Trainer Kit

Theory

INTERFACING REMOTE BOARD PERIPHERALS

ZigBee trainer peripherals are control by remotely also by using AT+REMOTE Command. For controlling other device peripherals at least two trainers required. In that one should be configured as Coordinator and other as Router or End Device. Refer ZigBee configuration section more about configuration.

The only difference between on board peripheral interface and remote board peripherals is that, we need to add an additional command AT+REMOTE other things are same as onboard peripherals interface.

1. INTERFACING WITH LEDS AND SWITCHS

To interfacing remote node peripherals we need at least two ZigBee trainers. One of them acts as coordinator other act as Router or End Device.

Below example show the complete operation of two ZigBee.

- Connect trainers to the PC and turn off all switches then Power On the trainer.
- Run ProBee Manger separately for two modules.
- Configure the Serial port (Baud - 9600, Data - 8, Parity - None, Stop Bit-1, Flow control -None).
- OPEN icon in ProBee manger is used to connect ZigBee module to PC.
- Press RESET TO DEFAULT ICON and reload the ZigBee using RELOAD icon.
- Go to Terminal window and follow the AT-Commands
- While sending data enable Local Echo.
- Enable the GPIO Pins as you required.

COORDINATOR	ROUTER
AT OK AT+NODENAME=COORDINATOR OK AT+NODETYPE=1 OK AT+PANID=ABCD OK ATS11=1 OK AT+GPIO= 2550002255550 OK ATZ OK OK # SET THE ROUTER ? AT+NODETYPE 1 OK AT+NODENAME? COORDINATOR OK	AT OK AT+NODENAME=ROUTER OK AT+NODETYPE=2 OK AT+PANID=ABCD OK ATS11=1 OK AT+GPIO= 2550002255550 OK ATZ OK OK # CHECK AT-CMD IN BOTH SIDE AT+NODETYPE? 2 OK AT+NODENAME? ROUTER OK
COORDINATOR	ROUTER
AT+PANID? ABCD OK AT+CHMASK? 03FFF000 OK AT+GPIO? 2550002255550 OK AT+PERMIT=255 OK AT+REMOTE=0001950000004D9E, AT+GPIO? 2550002255550 OK AT+REMOTE=0001950000004D9E, AT+DIO0=1 OK // remote node LED1 ON AT+REMOTE=0001950000004D9E, AT+DIO0=0 OK // remote node LED1 OFF AT+REMOTE=0001950000004D9E, AT+DIO10? 0 // remote node SW3 OFF (router) OK	AT+PANID? ABCD OK AT+CHMASK? 03FFF000 OK AT+GPIO? 2550002255550 OK AT+PERMIT=255 OK AT+LONGADDR? 0001950000004D9E OK Change the switch position to ON AT+REMOTE=0001950000004D95, AT+GPIO? 2550002255550 OK AT+REMOTE=0001950000004D95, AT+DIO0=1 OK // remote node LED1 ON AT+REMOTE=0001950000004D95, AT+DIO0=0

**AT+REMOTE=0001950000004D9E,
AT+DIO10?**

// remote node SW3 ON (router)
OK

**AT+LONGADDR?
0001950000004D95**
OK

Change the switch position to ON

OK // remote node LED1 OFF

**AT+REMOTE=0001950000004D95,
AT+DIO10?**

0 // remote node SW3 OFF (coordinator)
OK

**AT+REMOTE=0001950000004D95,
AT+DIO10?**

// remote node SW3 ON (coordinator)
OK

2. INTERFACING WITH RELAY & BUZZER.

To interfacing remote node peripherals we need at least two ZigBee trainers. One of them acts as coordinator other act as Router or End Device.

Below example show the complete operation of two ZigBee.

- Connect trainers to the PC and turn off all switches then Power On the trainer.
- Run ProBee Manger separately for two modules.
- Configure the Serial port (Baud - 9600, Data - 8, Parity - None, Stop Bit-1, Flow control -None).
- OPEN icon in ProBee manger is used to connect ZigBee module to PC.
- Press RESET TO DEFAULT ICON and reload the ZigBee using RELOAD icon.
- Go to Terminal window and follow the AT-Commands
- While sending data enable Local Echo.
- Enable the GPIO Pins as you required.

COORDINATOR	ROUTER
AT OK AT+NODENAME=COORDINATOR OK AT+NODETYPE=1 OK AT+PANID=ABCD OK ATS11=1 OK AT+GPIO= 5550005555522 OK ATZ OK OK ----- # SET THE ROUTER ? AT+NODETYPE 1 OK AT+NODENAME? COORDINATOR OK AT+PANID? ABCD OK AT+CHMASK? 03FFF000 OK AT+GPIO? 5550005555522 OK AT+PERMIT=255 OK	AT OK AT+NODENAME=ROUTER OK AT+NODETYPE=2 OK AT+PANID=ABCD OK ATS11=1 OK AT+GPIO= 5550005555522 OK ATZ OK OK ----- # CHECK AT-CMD IN BOTH SIDE AT+NODETYPE? 2 OK AT+NODENAME? ROUTER OK AT+PANID? ABCD OK AT+CHMASK? 03FFF000 OK AT+GPIO? 5550005555522 OK AT+PERMIT=255 OK

<pre> ----- AT+REMOTE=0001950000004D9E, AT+GPIO? 555000555522 OK AT+REMOTE=0001950000004D9E, AT+DIO11=1 OK // remote node BUZZER ON (router) AT+REMOTE=0001950000004D9E, AT+DIO11=0 OK // remote node BUZZER OFF (router) ----- </pre>	<pre> AT+LONGADDR? 0001950000004D9E OK ----- Change the switch position to ON ----- AT+REMOTE=0001950000004D95, AT+GPIO? 555000555522 OK AT+REMOTE=0001950000004D95, AT+DIO11=1 OK // remote node BUZZER ON (coordinator) </pre>
--	--

COORDINATOR	ROUTER
<pre> AT+REMOTE=0001950000004D9E, AT+DIO12=1 0 // remote node RELAY ON (router) OK AT+REMOTE=0001950000004D9E, AT+DIO12=0 // remote node RELAY OFF (router) OK ----- AT+LONGADDR? 0001950000004D95 OK ----- Change the switch position to ON ----- </pre>	<pre> AT+REMOTE=0001950000004D95, AT+DIO11=0 OK // BUZZER OFF (coordinator) ----- AT+REMOTE=0001950000004D95, AT+DIO12=1 0 // remote node RELAY ON (coordinator) OK AT+REMOTE=0001950000004D95, AT+DIO12=0 // remote node RELAY (coordinator) OK </pre>

3. INTERFACING WITH ADC USING VARIABLE RESISTOR.

To interfacing remote node peripherals we need at least two ZigBee trainers. One of them acts as coordinator other act as Router or End Device.

Below example show the complete operation of two ZigBee.

- Connect trainers to the PC and turn off all switches then Power On the trainer.
- Run ProBee Manger separately for two modules.
- Configure the Serial port (Baud - 9600, Data - 8, Parity - None, Stop Bit-1, Flow control -None).
- OPEN icon in ProBee manger is used to connect ZigBee module to PC.
- Press RESET TO DEFAULT ICON and reload the ZigBee using RELOAD icon.
- Go to Terminal window and follow the AT-Commands
- While sending data enable Local Echo.
- Enable the GPIO Pins as you required.

COORDINATOR	ROUTER
<pre># SET THE ROUTER AT+NODETYPE? 1 OK AT+NODENAME? COORDINATOR OK AT+PANID? ABCD OK AT+CHMASK? 03FFF000 OK AT+GPIO? 5550040055550 OK AT+PERMIT=255 OK ----- AT+REMOTE=0001950000004D9E, AT+GPIO? 5550040055550 OK AT+REMOTE=0001950000004D9E, AT+AI2? 2AF8 // remote node ADC Value OK AT+REMOTE=0001950000004D9E, AT+AI2? 0DA3 // remote node ADC Value OK ----- ++0001950000004D95 01100*0000000 ****,****,03DC,****</pre>	<pre># CHECK AT-CMD IN BOTH SIDE AT+NODETYPE? 2 OK AT+NODENAME? ROUTER OK AT+PANID? ABCD OK AT+CHMASK? 03FFF000 OK AT+GPIO? 5550040055550 OK AT+PERMIT=255 OK AT+LONGADDR? 0001950000004D9E OK ----- Change the position of variable res: ----- AT+REMOTE=0001950000004D95, AT+GPIO? 5550040055550 OK AT+REMOTE=0001950000004D95, AT+AI2? 0004 // remote node ADC Value (coordinator)</pre>

++0001950000004D95 01100*0000000 ****,****,03DC,****	OK

	Monitoring mode ROUTER

	AT+DESTLA=0001950000004D95
	OK
	ATS42=1
	OK
	ATS43=0
	OK
	ATZ
	OK
	OK

Monitoring mode CORDINATOR	Wait some time

ATS61=10	ATS42=0
OK	OK
ATZ	ATZ
OK	OK
OK	OK

Wait some time	

COORDINATOR	ROUTER
	<pre> ----- AT+DESTLA=0000000000000000 OK ATS42=1 OK ATS43=1 OK ----- ++0001950000004D95 01100*0000000 ****,****,03DC,**** ++0001950000004D95 01100*0000000 ****,****,03DC,**** ----- </pre>

4. INTERFACING WITH LDR

To interfacing remote node peripherals we need at least two ZigBee trainers. One of them acts as coordinator other act as Router or End Device.

Below example show the complete operation of two ZigBee.

- Connect trainers to the PC and turn off all switches then Power On the trainer.
- Run ProBee Manger separately for two modules.
- Configure the Serial port (Baud - 9600, Data - 8, Parity - None, Stop Bit-1, Flow control -None).
- OPEN icon in ProBee manger is used to connect ZigBee module to PC.
- Press RESET TO DEFAULT ICON and reload the ZigBee using RELOAD icon.
- Go to Terminal window and follow the AT-Commands
- While sending data enable Local Echo.
- Enable the GPIO Pins as you required.

COORDINATOR	ROUTER
AT OK AT+NODENAME=COORDINATOR OK AT+NODETYPE=1 OK AT+PANID=ABCD OK ATS11=1 OK AT+GPIO= 5550400055550 OK ATZ OK OK ----- -# SET THE ROUTER	AT OK AT+NODENAME=ROUTER OK AT+NODETYPE=2 OK AT+PANID=ABCD OK ATS11=1 OK AT+GPIO= 5550400055550 OK ATZ OK OK # CHECK AT-CMD IN BOTH SIDE AT+NODETYPE?

COORDINATOR	ROUTER
AT+NODETYPE? 1 OK AT+NODENAME? COORDINATOR OK AT+PANID? ABCD OK AT+CHMASK? 03FFF000 OK AT+GPIO? 5550400055550 OK AT+PERMIT=255 OK	2 OK AT+NODENAME? ROUTER OK AT+PANID? ABCD OK AT+CHMASK? 03FFF000 OK AT+GPIO? 5550400055550 OK AT+PERMIT=255 OK AT+LONGADDR? 0001950000004D9E OK
----- AT+REMOTE=0001950000004D9E, AT+GPIO? 5550400055550 OK AT+REMOTE=0001950000004D9E, AT+AI1? 2AF8 // remote node LDR Value OK AT+REMOTE=0001950000004D9E, AT+AI1? 0DA3 // remote node LDR Value OK	----- mask LDR ----- AT+REMOTE=0001950000004D95, AT+GPIO? 5550400055550 OK AT+REMOTE=0001950000004D95, AT+AI1? 0004 // remote node LDR Value (coordinator) OK
----- ++0001950000004D95 01100*0000000 ****,****,03DC,**** ++0001950000004D95 01100*0000000 ****,****,03DC,**** ----- Monitoring mode CORDINATOR ----- ATS61=10 OK ATZ OK OK ----- Wait some time -----	----- Monitoring mode ----- AT+DESTLA=0001950000004D95 OK ATS42=1 OK ATS43=0 OK ATZ OK OK ----- Wait some time ----- ATS42=0 OK ATZ OK OK

COORDINATOR	ROUTER
	<pre> ----- AT+DESTLA=0000000000000000 OK ATS42=1 OK ATS43=1 OK ----- ++0001950000004D95 01100*0000000 ****,****,03DC,**** ++0001950000004D95 01100*0000000 ****,****,03DC,**** ----- </pre>

5. INTERFACING WITH TEMPERATURE SENSOR

To interfacing remote node peripherals we need at least two ZigBee trainers. One of them acts as coordinator other act as Router or End Device.

Below example show the complete operation of two ZigBee.

- Connect trainers to the PC and turn off all switches then Power On the trainer.
- Run ProBee Manger separately for two modules.
- Configure the Serial port (Baud - 9600, Data - 8, Parity - None, Stop Bit-1, Flow control -None).
- OPEN icon in ProBee manger is used to connect ZigBee module to PC.
- Press RESET TO DEFAULT ICON and reload the ZigBee using RELOAD icon.
- Go to Terminal window and follow the AT-Commands
- While sending data enable Local Echo.
- Enable the GPIO Pins as you required.

COORDINATOR	ROUTER
<pre> AT OK AT+NODENAME=COORDINATOR OK AT+NODETYPE=1 OK AT+PANID=ABCD OK ATS11=1 OK AT+GPIO= 5554000055550 OK ATZ OK OK ----- -# SET THE ROUTER </pre>	<pre> AT OK AT+NODENAME=ROUTER OK AT+NODETYPE=2 OK AT+PANID=ABCD OK ATS11=1 OK AT+GPIO= 5554000055550 OK ATZ OK OK # CHECK AT-CMD IN BOTH SIDE AT+NODETYPE? </pre>

COORDINATOR	ROUTER
AT+NODETYPE? 1 OK AT+NODENAME? COORDINATOR OK AT+PANID? ABCD OK AT+CHMASK? 03FFF000 OK AT+GPIO? 5554000055550 OK AT+PERMIT=255 OK	2 OK AT+NODENAME? ROUTER OK AT+PANID? ABCD OK AT+CHMASK? 03FFF000 OK AT+GPIO? 5554000055550 OK AT+PERMIT=255 OK AT+LONGADDR? 0001950000004D9E OK
----- AT+REMOTE=0001950000004D9E, AT+GPIO? 5554000055550 OK AT+REMOTE=0001950000004D9E, AT+AI0? 2AF8 // remote node TMP Value OK AT+REMOTE=0001950000004D9E, AT+AI0? 0DA3 // remote node TMP Value OK	----- mask LDR ----- AT+REMOTE=0001950000004D95, AT+GPIO? 5554000055550 OK AT+REMOTE=0001950000004D95, AT+AI0? 0004 // remote node TMP Value (coordinator) OK
----- ++0001950000004D95 01100*0000000 ****,****,03DC,**** ++0001950000004D95 01100*0000000 ****,****,03DC,**** ----- Monitoring mode CORDINATOR ----- ATS61=10 OK ATZ OK OK ----- Wait some time -----	----- Monitoring mode ----- AT+DESTLA=0001950000004D95 OK ATS42=1 OK ATS43=0 OK ATZ OK OK ----- Wait some time ----- ATS42=0 OK ATZ OK OK

COORDINATOR	ROUTER
	<pre> ----- AT+DESTLA=0000000000000000 OK ATS42=1 OK ATS43=1 OK ----- ++0001950000004D95 01100*0000000 ****,****,03DC,**** ++0001950000004D95 01100*0000000 ****,****,03DC,**** ----- </pre>